Problem 1-

A polynomial is integral when it has integer coefficients. The square root of 2 is a solution to the integral polynomial equation $x^2-2=0$. A number is rational when it can be expressed as $\frac{a}{b}$ for integers a and b (with $b\neq 0$). A number is irrational when it is not rational.

a) suppose c is a non-square integer (that is, $c\neq n^2$ for any n). Explain why $\sqrt{c}$ is not rational. Similarly, if c is a non-cube integer, does it follow that $\sqrt[3]{c}$ is irrational?

~~suppose $\sqrt{c}=f$, Suppose $\sqrt{c}$ is rational~~

~~$f=(n+\frac{x}{y})$ $n\in\mathbb{Z}$, $y>x$~~

~~$(n+\frac{x}{y})(n+\frac{x}{y})=n^2+\frac{2nx}{y}+\frac{x^2}{y^2}=$~~

~~$n^2+\frac{2nx}{y}+\frac{x^2}{y^2}=$~~

~~if $\frac{2nx}{y}+\frac{x^2}{y^2}$ can't be an integer then $\sqrt{c}$ would be irrational~~

~~idea how through~~

Suppose $\sqrt{c}$ is rational and can be expressed as $\frac{x}{y}$ in its most simplified form, where $y\neq 0$

$(\frac{x}{y})^2=c$

$\frac{x}{y}\neq 1$ since then $(\frac{x}{y})^2=1$, hence $c=1$ and c cannot be a perfect square

since $\frac{x}{y} \neq 1$, $x \neq y$

$X \neq 0$ since if $x=0$

$\frac{x}{y} = 0$, $(\frac{x}{y})^2 = 0$, $C = 0$ and this is impossible since C cannot equal a square number

Suppose $x < y$:

$\frac{x^2}{y^2} < 1$ and $\frac{x^2}{y^2} > 0$ (since $x \neq 0$) (since squaring a fraction less than one makes it smaller)

there are no integers in the domain $(0, 1)$, and since C is an integer $x \not< y$

suppose $x > y$:

Since $\frac{x}{y}$ is a fraction in its simplest form, x and y share no factors. Hence x^2 and y^2 share no prime factors since squaring does not create new prime factors. $\frac{x^2}{y^2}$ won't simplify, hence $\frac{x^2}{y^2}$ cannot be an integer, and since C is an integer $x \not> y$. Since X cannot be greater than, less than, or equal to y, C cannot be expressed as $\frac{x}{y}$, hence C is irrational

ex $(\frac{3}{4})^2$ $\frac{9}{16}$ - prime factors: 3 → prime factors: 3

prime factors: 2 → prime factors: 2 no new prime factors created by squaring

*also $y \neq 1$ since then $\frac{x}{y}$ would itself be an integer, and then $(\frac{x}{y})^2$ would be a square number, and C can't be square

Finally, since the case of xcy didn't address negative numbers; negative numbers are not important to the question of is $\sqrt[3]{C}$ rational as if both numbers are negative it cancels out, and if one number is negative $\frac{x^2}{y^2}$ still yields the same result as it would if both x and y were positive since $(-1)^2=1$ and $1 \cdot x = x$

The same logic above that applies to squares applies to cubes, and so if C is a non-cube integer, it follows that $\sqrt[3]{C}$ is irrational.

The fact that it's a cube root rather than a square root does change the logic about negatives, however the signs of x and y are still trivial given that the concern is if $(\frac{x}{y})^3$ is an integer, not the actual value of $(\frac{x}{y})^3$. Alternatively you could apply the same logic with the squares, just with using absolute values/signs.

b) Find an integral polynomial equation that has $\alpha = \sqrt{3}+\sqrt{5}$ as a solution. Show that α is irrational

Let $x=\alpha$

$$[x-(\sqrt{3}+\sqrt{5})][x+(\sqrt{3}+\sqrt{5})]=0$$

$$x^2+x(\sqrt{3}+\sqrt{5})-x(\sqrt{3}+\sqrt{5})-(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})=0$$

doesn't cancel out

Leven if the 2nd and 3rd term cancel out the last term will create a root

$(x^2-3)(x^2-5)=0$

$x^4-5x^2-3x^2+15=0$

$x^4-8x^2+15=0$ doesn't work

$x=\sqrt{5}+\sqrt{3}$ also add roots of:

$x=-(\sqrt{5}+\sqrt{3})$ $x=-\sqrt{5}+\sqrt{3}$ $x=-\sqrt{3}+\sqrt{5}$

$[x-(\sqrt{5}+\sqrt{3})][x+(\sqrt{5}+\sqrt{3})][x+(\sqrt{3}-\sqrt{5})][x+(\sqrt{5}-\sqrt{3})]$

$[x^2+x(\sqrt{5}+\sqrt{3})-x(\sqrt{5}+\sqrt{3})-(\sqrt{5}+\sqrt{3})(\sqrt{5}+\sqrt{3})]$ left + two 0

$(x^2+5+3+2\sqrt{5}\sqrt{3})[x^2+x(\sqrt{5}-\sqrt{3})+x(\sqrt{3}-\sqrt{5})+(\sqrt{3}-\sqrt{5})(\sqrt{3}-\sqrt{5})]$

$(x^2+8+2\sqrt{5}\sqrt{3})(x^2+\sqrt{5}x-\sqrt{3}x+\sqrt{3}x-\sqrt{5}x+3+5-\sqrt{3}\sqrt{5}-\sqrt{3}\sqrt{5})$

$(x^2+8+2\sqrt{5}\sqrt{3})(x^2+8-2\sqrt{5}\sqrt{3})$

$x^4+8x^2-2\sqrt{5}\sqrt{3}x^2+8x^2+64-16\sqrt{5}\sqrt{3}+2\sqrt{5}\sqrt{3}x^2+16\sqrt{5}\sqrt{3}-4(5)(3)$

$x^4+16x^2+64-60$

$x^4+16x^2+4=0$

** = after having completed this page, I realize I made a sign mistake right here which likely caused my error. The star sign will indicate where I redo the problem starting from the plate at my mistake. I have also noticed a sign mistake here caused by a switched foot here. I will redo that part as well

Check:

$(\sqrt{3}+\sqrt{5})^4+16(\sqrt{3}+\sqrt{5})^2+4=0$

$(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})+16(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})+4=0$

$(8+2\sqrt{3}\sqrt{5})(8+2\sqrt{3}\sqrt{5})+16(8+2\sqrt{3}\sqrt{5})+4=0$

all positive, won't work

what if I change the above expression so that root terms cancel out, and then replace constant so the left side $=0$ $y\in\mathbb{Z}$

$(8+2\sqrt{3}\sqrt{5})(8+2\sqrt{3}\sqrt{5})-16(8+2\sqrt{3}\sqrt{5})+y=0$

$64+32\sqrt{3}\sqrt{5}+60-128-32\sqrt{3}\sqrt{5}+y=0$

$124-128+y=0$

$y=4$ → interesting, so original equation was all correct except the x^2 term sign of the + term

$x^4-16x^2+4=0$

Check:

$(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{3})-16(\sqrt{3}+\sqrt{3})(\sqrt{3}+\sqrt{5})+4=0$

$[8+2\sqrt{3}\sqrt{5}][8+2\sqrt{3}\sqrt{5}]-16(8+2\sqrt{3}\sqrt{5})+4=0$

$64+32\sqrt{3}\sqrt{5}+60-128-32\sqrt{3}\sqrt{5}+4=0$

$124-128+4=0$

$0=0$ ✓

$\alpha^4-16\alpha^2+4=0$ is an integral polynomial equation that has $\alpha=\sqrt{5}+\sqrt{3}$ as a solution

* $[x-(\sqrt{5}+\sqrt{3})][x+(\sqrt{5}+\sqrt{3})][x+(\sqrt{3}-\sqrt{5})][x+(\sqrt{5}-\sqrt{3})]$

$[x^2+x(\sqrt{5}+\sqrt{3})-x(\sqrt{5}+\sqrt{3})-(\sqrt{5}+\sqrt{3})(\sqrt{5}+\sqrt{3})](x^2+\sqrt{5}x-\sqrt{5}x+\sqrt{3}x+\sqrt{3}\sqrt{5}-3-\sqrt{5}x-5+\sqrt{3}\sqrt{5})$

$[x^2-(5+2\sqrt{3}\sqrt{5}+3)](x^2+2\sqrt{3}\sqrt{5}-8)$

$(x^2-8-2\sqrt{3}\sqrt{5})(x^2-8+2\sqrt{3}\sqrt{5})$

$x^4-8x^2+2\sqrt{3}\sqrt{5}x^2-8x^2+64-16\sqrt{3}\sqrt{5}-2\sqrt{3}\sqrt{5}x^2+16\sqrt{3}\sqrt{5}-4(3)(5)$

$x^4-16x^2+64-60$

x^4-16x^2+4

Show that α is irrational:

Originally I thought this could be achieved through proving that the sum of any two irrational numbers must be irrational, however I realized this is not necessarily the case

e.g. x = irrational number $= 0.241352\ldots$
y = irrational number $= \frac{2}{3} - x = 0.425314\ldots$

Instead $\sqrt{3}+\sqrt{5}$ can be shown to be irrational by viewing the quantity as a square root

$$(\sqrt{3}+\sqrt{5})^2$$
$$(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5}) =$$
$$3+5+2\sqrt{3}\sqrt{5} =$$
$$8+2\sqrt{15}$$

using the explanation from part a, I can now say $\sqrt{15}$ is irrational as it is a the square root of a non-square integer.

$2\sqrt{15}$ must also be irrational because if we assume $2\sqrt{15}$ is rational, then we can say it can be expressed as $\frac{x}{y}$, and therefore $\sqrt{15}$ can be expressed as $\frac{1}{2} \cdot \frac{x}{y}$, or $\frac{x}{2y}$, which is also rational. Since we have established $\sqrt{15}$ is irrational, $2\sqrt{5}$ must also be irrational.

$8+2\sqrt{5}$ must then also be irrational since if it was rational then $8+2\sqrt{15}$ could be written as $\frac{x}{y}$, and $2\sqrt{15}$ could therefore be written as $\frac{x}{y} - \frac{8}{1}$, which will be simplified to $\frac{x-8y}{y}$, which is a rational expression. However, since we have established that $2\sqrt{15}$ is irrational, $8+2\sqrt{15}$ must then also be irrational. Another way to think of this is that the addition of 8 only changes the digits before the decimal place in an already irrational number.

which in fact equals $\sqrt{3}+\sqrt{5}$

call this number B

To prove that $\sqrt{3}+\sqrt{5}$ is irrational, let's assume that the $\sqrt{8+2\sqrt{15}}$ is rational if this is true then B can be expressed as $\frac{x}{y}$. B^2 can then be expressed as $(\frac{x}{y})^2$ which equals $\frac{x^2}{y^2}$. B^2 also equals $8+2\sqrt{15}$ since B is defined as $\sqrt{8+2\sqrt{15}}$. Therefore $\frac{x^2}{y^2}=8+2\sqrt{15}$. However $8+2\sqrt{15}$ is irrational, and so it can't be expressed as a fraction. Therefore $\sqrt{8+2\sqrt{15}}$ must also be irrational. $\sqrt{8+2\sqrt{15}}=\sqrt{3}+\sqrt{5}$, so $\sqrt{3}+\sqrt{5}$ must be irrational.

Hence,

If $\alpha=\sqrt{3}+\sqrt{5}$, α must be irrational.

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c) Let a and b be integers. Find an integral polynomial equation which has $\sqrt{a}+\sqrt{b}$ as a solution. Must $\sqrt{a}+\sqrt{b}$ be irrational? if $a \neq b$, must $\sqrt{a}-\sqrt{b}$ be irrational?

First part can be approached like first part of part b.

$$[x-(\sqrt{a}+\sqrt{b})][x+(\sqrt{a}+\sqrt{b})][x+(\sqrt{a}-\sqrt{b})][x+(\sqrt{b}-\sqrt{a})]=0$$

$$[x^2+x(\sqrt{a}+\sqrt{b})-x(\sqrt{a}+\sqrt{b})-(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b})][x^2+\cancel{x\sqrt{b}}-\cancel{x\sqrt{a}}+\cancel{x\sqrt{a}}+\cancel{x\sqrt{b}}-a-\cancel{x\sqrt{b}}-b+\sqrt{a}\sqrt{b}]=0$$

$$(x^2-a-2\sqrt{a}\sqrt{b}-b)(x^2-a+2\sqrt{a}\sqrt{b}-b)=0$$

$$x^4-ax^2+2\cancel{\sqrt{a}\sqrt{b}x^2}-bx^2-ax^2+a^2-2\cancel{\sqrt{a}\sqrt{b}a}+ab-2\cancel{\sqrt{a}\sqrt{b}x^2}+2\cancel{\sqrt{a}\sqrt{b}a}-4ab+2\cancel{\sqrt{a}\sqrt{b}b}-bx^2+ba-2\cancel{\sqrt{a}\sqrt{b}b}+b^2=0$$

$$x^4-ax^2-bx^2-ax^2+a^2+ab-4ab-bx^2+ab+b^2=0$$

$$x^4-2ax^2-2bx^2+a^2+b^2-2ab=0$$

2(5)+2(3)=16, so when $\sqrt{5}+\sqrt{3}$ the coefficient of x^2 is 16, which it was in part b

Check:

$$(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b})-2a(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b})-2b(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b})+a^2+b^2-2ab=0$$

$$(a+2\sqrt{a}\sqrt{b}+b)(a+2\sqrt{a}\sqrt{b}+b)-2a(a+2\sqrt{a}\sqrt{b}+b)-2b(a+2\sqrt{a}\sqrt{b}+b)+a^2+b^2-2ab=0$$

$$a^2+2\sqrt{a}\sqrt{b}a+ab+2\sqrt{a}\sqrt{b}a+4ab+2\sqrt{a}\sqrt{b}b+ab+2\sqrt{a}\sqrt{b}b+b^2-2a^2-4\sqrt{a}\sqrt{b}a-2ab-2ab-4\sqrt{a}\sqrt{b}b-2b^2+a^2+b^2-2ab=0$$

$$0=0$$

must $\sqrt{a}+\sqrt{b}$ be irrational
again we can think of $(\sqrt{a}+\sqrt{b})^2$

$(\sqrt{a}+\sqrt{b})(\sqrt{a}+\sqrt{b}) = a+2\sqrt{a}\sqrt{b}+b$ → this part of fractions should be done because my original logic used domain restrictions (from 0 to 2) that no longer apply.

When $a+2\sqrt{a}\sqrt{b}+b$ is a non-square integer, $\sqrt{a}+\sqrt{b}$ must be irrational, as shown in part a. If $a+2\sqrt{a}\sqrt{b}+b$ is a non-square, rational, and not an integer, let that number be F and let $\sqrt{F}=D$

Assume D is rational. Let $\frac{x}{y}$ represent D

$x \neq 0$ because F is not an integer

$x \neq y$ because F is not an integer

$x \neq y$ since no new prime factors are created by squaring, and since $\frac{x}{y}$ is fully simplified, you will not be able to simplify $\frac{x^2}{y^2}$

wait $\frac{x^2}{y^2}$ not supposed to be equal to $\frac{x}{y}$ so I don't know how this would work

Since F is rational, let it be defined as $\frac{x^2}{y^2}$,
$D=\sqrt{F}=\sqrt{\frac{x^2}{y^2}}=\frac{\sqrt{x^2}}{\sqrt{y^2}}=\frac{x}{y}$. Hence, D must be rational
Wait, this logic doesn't work since x or y itself may be irrational

if One square number and One non-square integer, then the result is certainly irrational, as the result of the non-square integer must be irrational and the square-root of the square would be an integer, so the logic regarding why $\sqrt{5}$ being irrational implies $2\sqrt{5}$ being irrational applies

The question, then, is what if it's an irrational number divided by an irrational number

I ~~know~~ think my earlier attempt may work. Let $\frac{x}{y}$ is fully simplified
$F=\frac{x}{y}$ and let $D=\sqrt{F}$. F is not an integer. let x and the integers

$x \neq 0$ as F is not an integer
$x \neq y$ as F is not an integer
$x \not> y$

Suppose F is non-integer non-square rational number. $F=D^2$
Assume D is rational, and can be expressed in the fully simplified form $\frac{x}{y}$
F is then $\frac{x^2}{y^2}$

this is actually possible though, for example if $D=\frac{3}{5}$

Perhaps $\frac{\sqrt{a}}{\sqrt{b}}$ where a and b are non-square integers can be manipulated into a better form.

$\frac{\sqrt{3}}{\sqrt{5}} = \frac{\sqrt{15}}{5} = \frac{1}{5} \cdot \sqrt{15}$ $\frac{\sqrt{a}}{\sqrt{b}} = \frac{\sqrt{ab}}{b}$ pg 10

This solves the problem, since the same logic in part b may now be used.

now
I ~~knew~~ realize "non-squares" is redundant since I already say non-integers

Define F as a rational non-integer who can be most simply expressed as $\frac{x}{y}$ where x and y are integers.
Let $D = \sqrt{F}$
$D = \sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}} = \frac{\sqrt{x}\sqrt{y}}{y} = \frac{\sqrt{xy}}{y}$, since x and y are both integers xy must be an integer.

if xy is a square number, $\sqrt{xy}$ is rational, and so D is rational

if xy is not a square number, $\sqrt{xy}$ is not rational (as shown in part a) and so $\frac{\sqrt{xy}}{y}$ is not rational. This is because if $\frac{\sqrt{xy}}{y}$ was rational, it would be equivalent to $\frac{1}{y} \cdot \sqrt{xy}$, meaning $\sqrt{xy}$ would have to be rational and $\sqrt{xy}$ is not rational. This is the same logic that is used in part B. Hence $\frac{\sqrt{xy}}{y}$ is irrational

this was found through rationalizing the denominator in above steps

The only type of rational root for integers are squares, and the only type of rational roots for fractions (in form of $\frac{x}{y}$ where x and y are integers, though every fraction should be able to be put in this form are ones where the numerator and denominator multiply to a square number.

actually this is unnecessary since square times 1 is a square

So if $a + 2\sqrt{a}\sqrt{b} + b$ is a square number or a fraction whose numerator and denominator multiply to a square $\sqrt{a} + \sqrt{b}$ will be rational

it seems any time you have $\frac{a}{b}$ and $ab=n^2$ $n\in\mathbb{Z}$ pg 11 $\frac{a}{b}$ can be simplified to a square over a square, which makes sense considering both have rational solutions

Obviously if a and b are squares, or their numerator and denominator multiply to a square (as seen on last page) $\sqrt{a}+\sqrt{b}$ is rational. Are there any cases where this is not the case, yet $\sqrt{a}+\sqrt{b}$ is still rational?

what if we set $a=\frac{3}{5}$ and $a+2\sqrt{a}\sqrt{b}+b=9$

$$\frac{3}{5}+2\sqrt{\frac{3}{5}}\sqrt{b}+b=9$$
$$b+2\sqrt{\frac{3}{5}}\sqrt{b}=\frac{42}{5}$$
$$b-\frac{42}{5}=-2\sqrt{\frac{3}{5}}\sqrt{b}$$
$$b^2-\frac{84}{5}b+\frac{1764}{25}=(4)\left(\frac{3}{5}\right)b$$
$$b^2-\frac{84}{5}b+\frac{1764}{25}=\frac{12}{5}b$$
$$b^2-\frac{96}{5}b+\frac{1764}{25}=0$$
$$b=\frac{\frac{96}{5}\pm\sqrt{\frac{9216}{25}-\frac{7056}{25}}}{2}$$

negative number

$b=\frac{\frac{96}{5}-12\sqrt{15}}{2}\approx 1.3639$

$b=\frac{\frac{96}{5}+12\sqrt{15}}{2}$

$b=\frac{\frac{96}{5}+12\sqrt{15}}{2}\approx 32.837$

greater than 9

above is also tangential as a should be an integer

$a=5$

$a+2\sqrt{a}\sqrt{b}+b=16$

$5+2\sqrt{5}\sqrt{b}+b=16$

$b+2\sqrt{5}\sqrt{b}=11$

$b-11=-2\sqrt{5}\sqrt{b}$

$b^2-22b+121=(4)(5)(b)$

$b^2-22b+121=20b$

$b^2-42b+121=0$

$b=\frac{42\pm\sqrt{1764-484}}{2}$

$b=\frac{42\pm\sqrt{1280}}{2}$

works

$b=\frac{42+16\sqrt{5}}{2}$

$b=\frac{42-16\sqrt{5}}{2}$

≈ 38.88

≈ 3.11

however, b is not an integer in this case

b also included $\sqrt{5}$ and a was 5, which could never happen with 2 distinct integers

for example $\sqrt{3}+\sqrt{5}$

$(\sqrt{3}+\sqrt{5})(\sqrt{3}+\sqrt{5}) = 8+2\sqrt{15}$

$\sqrt{12}+\sqrt{3}$

$(\sqrt{12}+\sqrt{3})(\sqrt{12}+\sqrt{3}) = 15+2\sqrt{36} = 15+12 = 27$

$\sqrt{20}+\sqrt{5} = 4\sqrt{5}+\sqrt{5} = 5\sqrt{5}$

$\sqrt{a}+\sqrt{5}$ $\quad a \in \mathbb{Z}$ $\quad b \in \mathbb{Z}$

take the square of them

$a + 2\sqrt{a}\sqrt{b} + b$

a and b are integers, so a+b is an integer

$2\sqrt{a}\sqrt{5}$

if a [redacted] is a square number and b is not, $\sqrt{a}\sqrt{5}$ will not be square, and hence irrational (part A)

Finally, if $a+b+2\sqrt{a}\sqrt{b}$ is irrational $\sqrt{a}+\sqrt{b}$ is irrational, because if it were rational it could be expressed as $\frac{x}{y}$, and so $a+b+2\sqrt{a}\sqrt{b}$ could be expressed as $\frac{x^2}{y^2}$, which must be false because $a+b+2\sqrt{a}\sqrt{b}$ is irrational

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the reason the product of a square and non-square number must be a non-square number is because each factor in the square number already has a repeat, since it's a square, and at least one factor in the non-square number is missing a repeat factor, since it's not a square. Each factor in the square is already set together with it's pair, so there is no extra for the non-repeating factor

ex) $\sqrt{9}\cdot\sqrt{8} = \sqrt{3\cdot3\cdot(2\cdot2)\cdot2}$

(3·3: repeated factor; 2: original unpaired factor can't find a pair)

Any time $a=n^2$ and $b\neq n^2$, $\sqrt{a}+\sqrt{b}$ is irrational because you will end up with an integer $(a+b)$ plus 2 times $\sqrt{C}$, where C is a non-square integer, which means it's irrational, and we can say $2\sqrt{C}$ is irrational for the same reason shown in part B, because if $2\sqrt{C}$ was rational it could be expressed as $\frac{x}{y}$, meaning $\sqrt{C}$ would be $\frac{x}{2y}$, but since $\sqrt{C}$ is irrational this can't be true. An integer plus an irrational number must be irrational as an integer can be expressed as $\frac{z}{1}$, and if the sum were rational it could be expressed as $\frac{x}{y}$, that would mean $2\sqrt{C}$ could be expressed as $\frac{x}{y}-\frac{z}{1}\rightarrow\frac{x-zy}{y}$ and since $\frac{x}{y}$ is in simplest form x, y, and z are all integers, so $\frac{x-zy}{y}$ is rational, and since $2\sqrt{C}$ is irrational this must be false.

in simplest form

~~an integer since A and B are integers and so also 2√n~~

The next case to look at is $\sqrt{a}+\sqrt{b}$ where pg 14
$a \neq n^2$ $b \neq n^2$ $n \in \mathbb{Z}$

the last page shows that if $\sqrt{a}\sqrt{b}$ is irrational
$\sqrt{a}+\sqrt{b}$ is irrational (assuming a and b are integers)

However there are some cases that fulfill the two above requirements

unpartnered factor

ex $\sqrt{3}\cdot\sqrt{12} = \sqrt{3}\cdot\sqrt{2}\sqrt{2}\sqrt{3} = \sqrt{3}\sqrt{3}\cdot\sqrt{2}\sqrt{2} = 3\cdot 2 = 6$

unpartnered factor — repeated factor

There are two cases where $\sqrt{a}\sqrt{b} = n^2$ $n \in \mathbb{Z}$

one is where a and b are squares, which is not the current case

The other is where two a singular square pair. There also may be other terms that are already paired, such as in $\sqrt{12}$, however they are irrelevant as they are square numbers and can be pulled out of $\sqrt{a}\sqrt{b}$ while still making the coefficient of $\sqrt{a}\sqrt{b}$ an integer, when it comes to $\sqrt{a}+\sqrt{b}$ the same thing applies as the square numbers can be taken out of the root and turned into coefficients

This means $\sqrt{a} + \sqrt{b}$ can be rewritten as $d\sqrt{a_2} + f\sqrt{b_2}$ where $f \in \mathbb{Z}$ $d \in \mathbb{Z}$

Since we're dealing with a case where $\sqrt{a}\sqrt{b}$ is an integer, a_2 and b_2 are equal, since they are the unpaired factors. Hence $d\sqrt{a_2} + f\sqrt{b_2} = d\sqrt{a_2} + f\sqrt{a_2} =$

$(d+f)\sqrt{a_2}$. a_2 is a non-square integer since the square term (could be 1, if no square term) was taken out and the factors are integers (you don't take out a square if it leaves you with a fraction), and since the square root of a non-square integer is irrational (part a), $\sqrt{a_2}$ must be irrational, since d and f are integers (square numbers that are square rooted produce integers) d+f is an integer (all sums of integers are integers), and an integer times an irrational number is irrational, as shown in part B and on the last page. Therefore $\sqrt{a} + \sqrt{b}$ where $a \in \mathbb{Z}$ and $a \neq n^2$ and where $b \in \mathbb{Z}$ and $b \neq n^2$ must be irrational

ex $\sqrt{12} + \sqrt{3} = \sqrt{4}\sqrt{3} + \sqrt{3} = 2\sqrt{3} + \sqrt{3} = 3\sqrt{3}$ → irrational (irrational)

Finally there is the case of $\sqrt{a} + \sqrt{b}$, $a \in \mathbb{Z}$ and $a = n^2$, $b \in \mathbb{Z}$ and $b = n^2$

Since the definition of a square number is one whose square root is an integer, a and b are integers, ~~Hence a+b can be expressed~~ Integers already have a rational numerator and a common denominator, the sum of integers must be rational

Hence $\sqrt{a}+\sqrt{b}$ where $a \in \mathbb{Z}$ and $a=n^2$ and $b \in \mathbb{Z}$ and $b=n^2$ is rational

~~$\sqrt{a}+\sqrt{b}$ is rational if and only if~~

$\sqrt{a}+\sqrt{b}$, where a and b are integers, is rational iff (if and only if) a and b are square numbers ($a=n^2, b=n^2, n \in \mathbb{Z}$)

The same exact logic may be applied to $\sqrt{a}-\sqrt{b}$ when $a \neq b$ (else the term (b+f) in the previous explanation becomes 0, and 0 times anything is 0, which is rational), meaning $\sqrt{a}-\sqrt{b}$ $a \neq b$ $a \in \mathbb{Z}$, $b \in \mathbb{Z}$ is rational iff $a=n^2 n \in \mathbb{Z}$, $b=n^2 n \in \mathbb{Z}$

I just realized there's a much easier way to prove the case of $\sqrt{a}+\sqrt{b}$ where $a=n^2$ and $b \neq n^2$ $\sqrt{b}$ is irrational (part a) and $\sqrt{a}$ is an integer, if $\sqrt{a}+\sqrt{b}$ was rational it could be written as $\frac{x}{y}$ in reduced form so $\sqrt{b}$ could be written as $\frac{x}{y}-\frac{z}{1} \rightarrow \frac{x-zy}{y}$, and x, y, and z are integers (x and y because $\frac{x}{y}$ is in reduced form, z because $\sqrt{a}$ is an integer) and so $\frac{x-zy}{y}$ is rational, which would mean $\sqrt{b}$ is rational, which is false. Hence $\sqrt{a}+\sqrt{b}$ is irrational

d) Formulate some generalizations. As a starting point, pg 17
is $B=\sqrt{3}+\sqrt{5}+\sqrt{7}$ irrational? What about numbers like
$\gamma=3\sqrt{2}-2\sqrt{3}-3\sqrt{5}+\sqrt{6}$ and $\delta=\sqrt{5}-\sqrt{2}$

$(\sqrt{3}+\sqrt{5}+\sqrt{7})^2$

$(\sqrt{3}+\sqrt{5}+\sqrt{7})(\sqrt{3}+\sqrt{5}+\sqrt{7}) = 3+\sqrt{3}\sqrt{5}+\sqrt{3}\sqrt{7}+\sqrt{5}\sqrt{5}+5+\sqrt{5}\sqrt{7}+\sqrt{3}\sqrt{7}+\sqrt{5}\sqrt{7}+7$
$3+5+7+2\sqrt{3}\sqrt{5}+2\sqrt{3}\sqrt{7}+2\sqrt{5}\sqrt{7}$

if none of the the terms are squares then everything in the square root is either non-square, which would mean irrationally or a repeated root, which would mean those roots would be like terms and their sum would be irrational. The question is how to evaluate if different non square integers are rooted and then added, must that some be irrational. There's a semi-compelling case to say that $\frac{x}{y}+\frac{A}{B}=\frac{xB+Ay}{yB}$, which is rational, so it that's rational the first two terms must be rational, however there is alway the case of $a=0.51234\ldots$ $b=\frac{2}{3}-a=0.15432\ldots$
→ although this some number is defined by not a square root of a different number

However in this case we know the square rooted terms are integers, let's say $\sqrt{3}+\sqrt{5}=\frac{x}{y}$

$\frac{x}{y}-\sqrt{5}=\sqrt{3}$
$(\frac{x}{y}-\sqrt{5})^2=3$
$\frac{x^2}{y^2}-2\sqrt{5}\frac{x}{y}+5=3$
→ $-2-\frac{x^2}{y^2}=-2\sqrt{5}\frac{x}{y}$
$\frac{x^2}{y^2}+2=2\sqrt{5}\frac{x}{y}$
$\frac{x^2y}{y^2x}+2=2\sqrt{5}$
$\sqrt{5}=\frac{1}{2}(\frac{x^2y}{y^2x}+2)$
$\sqrt{5}=\frac{1}{2}\left(\frac{x^2y+2y^2x}{y^2x}\right)$
$\sqrt{5}=\frac{x^2y+2y^2x}{2y^2x}$
$\sqrt{5}$ must be rational, but it isn't, so $\sqrt{3}+\sqrt{5}$ must be irrational

The next step is to try to generalize this for the sum of any non-square integer roots.

$\sqrt{a}+\sqrt{b}=\frac{x}{y}$ → reduced form of fraction (expressed in integers)

$\sqrt{a}=\frac{x}{y}-\sqrt{b}$

$a=\frac{x^2}{y^2}-2\frac{x}{y}\sqrt{b}+b$

$a-b-\frac{x^2}{y^2}=-2\frac{x}{y}\sqrt{b}$

$2\frac{x}{y}\sqrt{b}=b-a+\frac{x^2}{y^2}$

$2\sqrt{b}=\frac{by}{x}-\frac{ay}{x}+\frac{x^3y}{xy^2}$

$\sqrt{b}=\left(\frac{by}{x}-\frac{ay}{x}+\frac{x^2y}{xy^2}\right)\frac{1}{2}$

$\sqrt{b}=\left(\frac{by^3-ay^3+x^2y}{xy^2}\right)\frac{1}{2}$

$\sqrt{b}=\frac{by^3-ay^3+x^2y}{2xy^2}$

b is an integer (assumption, scope of proof), a is an integer (same reason as b), x is an integer and so is y, since $\frac{x}{y}$ is reduced form. 2 is an integer (by definition). Since all the terms set equal to $\sqrt{b}$ are integers, $\sqrt{b}$ must be rational. Since $\sqrt{b}$ is irrational (b is a non-square integer), $\sqrt{a}+\sqrt{b}$ must be irrational.

Any square number square rooted is an integer, and an integer plus an irrational number is irrational, so $\sqrt{a}+\sqrt{b}$ must be irrational if one of them is irrational

This is the same conclusion as from part C, however this proof is much more useful, as it doesn't rely on looking at the squares of combinations and can be applied to n terms as you can always isolate one of them and write them in terms of the other, and you can use common denominators to only remain with 1 term. So if you have $\sqrt{a}+\sqrt{b}+\sqrt{c}\ldots$ as long as one of them is irrational, the summation must also be irrational. This is also true of cases with coefficients in front of terms, provided the coefficients themselves aren't irrational (such as $\sqrt{3}\sqrt{5}$)

ex) $\sqrt{a}+\sqrt{b}+\sqrt{c}=\frac{x}{y}$

$(\frac{x}{y}-\sqrt{b}-\sqrt{c})(\frac{x}{y}-\sqrt{b}-\sqrt{c})$

$\sqrt{a}=\frac{x}{y}-\sqrt{b}-\sqrt{c}$

$a=\frac{x^2}{y^2}-\frac{x}{y}\sqrt{b}-\frac{x}{y}\sqrt{c}-\frac{x}{y}\sqrt{b}+b+\sqrt{b}\sqrt{c}-\frac{x}{y}\sqrt{c}+\sqrt{b}\sqrt{c}+c$

$a=\frac{x^2}{y^2}-2\frac{x}{y}\sqrt{b}-2\frac{x}{y}\sqrt{c}+2\sqrt{b}\sqrt{c}+b+c$

$a-b-c-\frac{x^2}{y^2}=-2\frac{x}{y}\sqrt{b}-2\frac{x}{y}\sqrt{c}+2\sqrt{b}\sqrt{c}$

$\frac{x^2}{y^2}+b+c-a=2\frac{x}{y}\sqrt{b}+2\frac{x}{y}\sqrt{c}+2\sqrt{b}\sqrt{c}$

Maybe there is actually a problem with doing this.

What if I rewrite $\sqrt{5}$ as $\sqrt[4]{5^2}$, ~~because~~ actually this couldn't work as $\sqrt{5^2}$ can be rational

$$\frac{x^2}{y^2}+b+c-a=2\frac{x}{y}\sqrt{5}+2\frac{x}{y}\sqrt{c}+2\sqrt{5}\sqrt{c}$$

I will try using actual numbers first again

$$\sqrt{3}+\sqrt{5}+\sqrt{7}=\frac{x}{y}$$
$$\sqrt{3}+\sqrt{5}=\frac{x}{y}-\sqrt{7}$$
$$3+5+\sqrt{3}\sqrt{5}=\frac{x^2}{y^2}-2\frac{x}{y}\sqrt{7}+7$$
$$1+\sqrt{3}\sqrt{5}=\frac{x^2}{y^2}-2\frac{x}{y}\sqrt{7}$$
$$1+\sqrt{15}=\frac{x^2}{y^2}-2\frac{x}{y}\sqrt{7}$$
$$\sqrt{15}+2\frac{x}{y}\sqrt{7}=\frac{x^2}{y^2}-1$$

$$15+4\frac{x^2}{y^2}\cdot 7+4\frac{x}{y}\sqrt{7}\sqrt{15}=\frac{x^4}{y^4}-2\frac{x^2}{y^2}+1$$

$$4\frac{x}{y}\sqrt{105}=\frac{x^4}{y^4}-2\frac{x^2}{y^2}-28\frac{x^2}{y^2}-14$$

$$\sqrt{105}=\frac{1}{4}\left(\frac{x^4y}{xy^4}-\frac{30x^2y}{xy^2}-\frac{14y}{x}\right)$$

$$\sqrt{105}=\frac{1}{4}\left(\frac{x^4y-30x^2y^3-14y^5}{xy^4}\right)$$

$$\sqrt{105}=\frac{x^4y-30x^2y^3-14y^5}{4xy^4}$$

$\sqrt{a}\sqrt{b}\sqrt{c}$ is rational since a, b, and c are integers, that means a·b·c is a square number

That definitely means if a.b.c.d... is not a square number, the sum of their square roots are irrational (you can always combine the square root and isolate)

if $\sqrt{abc}$ is square it could mean a, b and c are all square numbers, and this would result in a rational solution for the sum of $\sqrt{a}+\sqrt{b}+\sqrt{c}$

it also could be a case of matching singular roots

$\sqrt{6}\cdot\sqrt{2}\cdot\sqrt{3} = \sqrt{2}\sqrt{2}\cdot\sqrt{3}\sqrt{3} = 2\cdot 3 = 6$

~~However then we can group the terms and simplify~~

$\sqrt{2}\sqrt{3}+\sqrt{2}+\sqrt{3}$ $\sqrt{6}+\sqrt{2}+\sqrt{3}$

$\sqrt{2}(\sqrt{3}+1)+\sqrt{3}$

$(\sqrt{2}\sqrt{3}+\sqrt{2}+\sqrt{3})(\sqrt{2}\sqrt{3}+\sqrt{2}+\sqrt{3})$

$6+2+3+2\sqrt{2}\sqrt{2}\sqrt{3}+2\sqrt{2}\sqrt{3}\sqrt{3}+2\sqrt{2}\sqrt{3}$

$11+4\sqrt{3}+6\sqrt{2}+2\sqrt{2}\sqrt{3}$

$(\sqrt{6}+\sqrt{2}+\sqrt{3})(\sqrt{6}+\sqrt{2}+\sqrt{3})(\sqrt{6}+\sqrt{2}+\sqrt{3})$ interesting how all roots that became integers were 12

$(11+4\sqrt{3}+6\sqrt{2}+2\sqrt{6})(\sqrt{6}+\sqrt{2}+\sqrt{3})$

$11\sqrt{6}+11\sqrt{2}+11\sqrt{3}+4\sqrt{18}+4\sqrt{6}+12+6\sqrt{12}+12+6\sqrt{6}+12+2\sqrt{12}+2\sqrt{18}$

$11\sqrt{6}+11\sqrt{2}+11\sqrt{3}+6\sqrt{18}+10\sqrt{6}+8\sqrt{12}+36$ $36+21\sqrt{6}+27\sqrt{3}+21\sqrt{2}$

~~$3(\quad)\sqrt{6}+$~~ $11\sqrt{6}+11\sqrt{2}+11\sqrt{3}+18\sqrt{2}+10\sqrt{6}+16\sqrt{6}+36$

maybe this can be proved generally

$\sqrt{a}\sqrt{b}+\sqrt{a}+\sqrt{b}=\frac{x}{y}$ → reduced fraction

$\sqrt{a}\sqrt{b}-\frac{x}{y}=\sqrt{a}+\sqrt{b}$

$ab+\frac{x^2}{y^2}-2\sqrt{a}\sqrt{b}=a+b+2\sqrt{a}\sqrt{b}$

$4\sqrt{a}\sqrt{b}=ab-a-b-\frac{x^2}{y^2}$

$\sqrt{a}\sqrt{b}=\frac{1}{4}(ab-a-b-\frac{x^2}{y^2})$

$\sqrt{a}\sqrt{b}=\frac{1}{4}\left(\frac{aby^2-ay^2-by^2-x^2}{y^2}\right)$

$\sqrt{a}\sqrt{b}=\frac{aby^2-ay^2-by^2-x^2}{4y^2}$

$\sqrt{2}\sqrt{3}+\sqrt{3}\sqrt{5}+\sqrt{2}\sqrt{3}\sqrt{5}+\sqrt{3}$

since a, b, x and y are integers $\sqrt{a}\sqrt{b}$ is rational, and $\sqrt{a}\sqrt{b}$ must be a square number (part a)

For any three square roots of integers being added, if one is irrational, the sum is irrational

Using the fact that the sum of three must be irrational the process could be done infinitely if the same logic works, just with setting larger terms equal to rational numbers

$\sqrt{6}+\sqrt{15}+\sqrt{30}+\sqrt{?}$

$(\sqrt{a}\sqrt{b}+\sqrt{b}\sqrt{c}+\sqrt{b})(\sqrt{a}\sqrt{b}+\sqrt{b}\sqrt{c}+\sqrt{b})$

$\sqrt{a}\sqrt{b}+\sqrt{b}\sqrt{c}+\sqrt{ab}\sqrt{b}\sqrt{c}+\sqrt{b}=\frac{x}{y}$

$\sqrt{a}\sqrt{b}\sqrt{c}-\frac{x}{y}=\sqrt{a}\sqrt{b}+\sqrt{b}\sqrt{c}+\sqrt{b}$

$abc+\frac{x^2}{y^2}-2\sqrt{a}\sqrt{b}\sqrt{c}=ab+b\sqrt{a}\sqrt{c}+b\sqrt{a}+b\sqrt{a}\sqrt{c}+bc+b\sqrt{c}+b\sqrt{a}+b\sqrt{c}+b$

$abc+\frac{x^2}{y^2}-2\sqrt{a}\sqrt{b}\sqrt{c}=ab+2b\sqrt{a}\sqrt{c}+2b\sqrt{a}+2b\sqrt{c}+b$

$-2\sqrt{a}\sqrt{b}\sqrt{c}-2b\sqrt{a}-2b\sqrt{c}-2b\sqrt{a}\sqrt{c}=ab+b-abc-\frac{x^2}{y^2}$

$\sqrt{a}\sqrt{b}\sqrt{c}+b\sqrt{a}+b\sqrt{c}+b\sqrt{ac}=\frac{-1}{2}(ab+b-abc-\frac{x^2}{y^2})$

$\frac{\sqrt{abc}}{b}+\sqrt{a}+\sqrt{c}+\sqrt{ac}=\frac{-1}{2b}(ab+b-abc-\frac{x^2}{y^2})$

$\frac{\sqrt{b}}{b}+\frac{1}{\sqrt{c}}+\sqrt{a}+1=\frac{-1}{2b}(ab+b-abc-\frac{x^2}{y^2})$

$\sqrt{ac}$

if it was something like where they paired up like
$\sqrt{2}+\sqrt{15}+\sqrt{10}+\sqrt{10} = 2\sqrt{15}+2\sqrt{10} = \sqrt{60}+\sqrt{40}$ → at least one non-square
$\sqrt{60}+\sqrt{40}$ must be irrational

$\frac{\sqrt{b}}{b}+\frac{1}{\sqrt{c}}+\frac{1}{\sqrt{a}}+1 = \frac{-\frac{1}{2b}(ab+a-abc-\frac{x^2}{b^2})}{\sqrt{ac}}$

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$\left(\frac{\sqrt{b}}{b}+\frac{1}{\sqrt{c}}+\frac{1}{\sqrt{a}}+1\right)\left(\frac{\sqrt{b}}{b}+\frac{1}{\sqrt{c}}+\frac{1}{\sqrt{a}}+1\right) = \frac{\frac{1}{4b^2}(ab+a-abc-\frac{x^2}{b^2})^2}{ac}$

$\frac{b}{b^2}+\frac{\sqrt{b}}{\sqrt{ac}}+\frac{\sqrt{b}}{\sqrt{a}}+\frac{\sqrt{b}}{b}+\frac{\sqrt{b}}{b\sqrt{c}}+\frac{1}{c}+\frac{1}{\sqrt{ac}}+\frac{1}{\sqrt{c}}+\frac{\sqrt{b}}{\sqrt{a}b}+\frac{1}{\sqrt{ac}}+\frac{1}{a}+\frac{1}{\sqrt{a}}+\frac{\sqrt{b}}{b}+\frac{1}{\sqrt{c}}+\frac{1}{\sqrt{a}}+1 =$

$\frac{b+2b\sqrt{b}}{b^2}+\frac{2+\sqrt{b}}{\sqrt{c}}+\frac{2+\sqrt{b}}{\sqrt{a}}+\frac{2}{\sqrt{ac}}+\frac{\sqrt{b}}{b\sqrt{c}}+\frac{1}{c}+\frac{1}{a}+\frac{\sqrt{b}}{\sqrt{a}b}+1 = \ldots$

$\frac{b+2b\sqrt{b}}{b^2}+\frac{2\sqrt{a}+\sqrt{ab}+2\sqrt{c}+\sqrt{bc}+2}{\sqrt{ac}}+\frac{\sqrt{ab}+\sqrt{bc}}{b\sqrt{ac}}+\frac{1}{c}+\frac{1}{a}+1 = \ldots$

$\frac{b\sqrt{ac}+2b\sqrt{abc}+2b\sqrt{a}+b^2\sqrt{ab}+b^2\sqrt{c}+b^2\sqrt{bc}+2b^2+b\sqrt{ab}+b\sqrt{bc}}{b^2\sqrt{ac}}+\frac{1}{c}+\frac{1}{a}+1 = \ldots$

$\ldots = \frac{\frac{1}{4b^2}(ab+a-abc-\frac{x^2}{b^2})^2-ac-ac}{ac}$

idk

$\frac{\sqrt{2}\sqrt{3}+\sqrt{3}\sqrt{3}+\sqrt{6}\sqrt{3}+\sqrt{3}}{(\sqrt{2}+\sqrt{6})(\sqrt{3}+(\sqrt{3}+1))(\sqrt{3})}$

$\sqrt{a}\sqrt{b}+\sqrt{b}\sqrt{c}+\sqrt{a}\sqrt{b}\sqrt{c}+\sqrt{b} = \frac{x}{y}$

$\sqrt{b}(\sqrt{a}+\sqrt{c}+\sqrt{ac}+1) = \frac{x}{y}$

$b(\sqrt{a}+\sqrt{c}+\sqrt{ac}+1) = \frac{x\sqrt{b}}{y}$

$b(\sqrt{a}+\sqrt{c}+\sqrt{ac}) = \frac{x\sqrt{b}-by}{y}$

$by(\sqrt{a}+\sqrt{c}+\sqrt{ac}) = x\sqrt{b}-by$

$b^2y^2(a+c+ac+\sqrt{ac}+a\sqrt{c}+\sqrt{ac}+(\sqrt{a}+a\sqrt{c}+(\sqrt{a}) = x^2b+b^2y^2-2xby\sqrt{b}$

$b^2y^2(a+c+ac+2\sqrt{ac}+2a\sqrt{c}+2(\sqrt{a}) = x^2b+b^2y^2-2xby\sqrt{b}$

$\sqrt{a}+\sqrt{b}+\sqrt{c}+\sqrt{d}$ $\quad a,b,c,d \in \mathbb{Z}$

$\sqrt{abcd}$ must be square, actually must be [redacted] since all numbers are integers

product of 2 squares

$\sqrt{6}+\sqrt{6}+\sqrt{30}+\sqrt{3}$

$\sqrt{2}\sqrt{3}+\sqrt{3}\sqrt{3}+\sqrt{6}\sqrt{3}+\sqrt{3}$

$(\sqrt{2}+\sqrt{3}+\sqrt{6})(\sqrt{3})$

numbers such as $\sqrt{3}+\sqrt{5}+\sqrt{7}$ and $3\sqrt{2}-2\sqrt{3}-3\sqrt{5}+\sqrt{6}$ are irrational (integer coefficients times irrational is still irrational)

Can rules such as $\sqrt{a}+\sqrt{5}$ is irrational apply to rational non-integers

I believe its the same idea, except $\sqrt{a}$ is rational if $\frac{x}{y}=a$ and $xy=n^2$ $n\in\mathbb{Z}$

$\sqrt{\frac{3}{4}}=\sqrt{\frac{1}{4}}\sqrt{3}=\frac{1}{2}\sqrt{3}$ → integers are just special cases of this rule where $y=1$ so the numerator must be square

$\sqrt{\frac{3}{5}}=\sqrt{\frac{1}{5}}\sqrt{3}=\frac{1}{\sqrt{5}}\cdot\sqrt{3}$
$=\sqrt{4}\sqrt{\frac{1}{6\frac{2}{3}}}$

Assume $a=\frac{x}{y}$, $a\in\mathbb{R}$ → x and y are both integers, $xy\neq n^2$ → in simplest form

$X\neq y$ because then $xy=x^2(xy\neq n^2)$

if $X>y$ $\sqrt{\frac{x}{y}}\left(\frac{\sqrt{x}}{\sqrt{x}}\right)=\sqrt{\frac{x^2}{xy}}=\frac{x}{\sqrt{xy}}$

if $x<y$ $\sqrt{\frac{x}{y}}\left(\frac{\sqrt{x}}{\sqrt{x}}\right)=\sqrt{\frac{x^2}{xy}}=\frac{x}{\sqrt{xy}}$

x and y are both integers, and there product is a non-square integer, so $\sqrt{xy}$ is irrational, so $\frac{x}{\sqrt{xy}}$ is irrational since any integer divided by an irrational number is irrational.

If $\frac{x}{\sqrt{xy}}$ could be written as $\frac{A}{B}$ then

$$\frac{x}{\sqrt{xy}} = \frac{A}{B} \rightarrow A\sqrt{xy} = Bx \quad \sqrt{xy} = \frac{Bx}{A}$$

$\sqrt{xy}$ could be expressed as $\frac{Bx}{A}$, where B, A, and x are all integers, making $\sqrt{xy}$ rational. Since $\sqrt{xy}$ is irrational, $\frac{x}{\sqrt{xy}}$ is also irrational

$$\sqrt[3]{5} - \sqrt{2}$$

$$\sqrt[3]{a} - \sqrt{b} = \frac{x}{y}$$

$$\sqrt[3]{a} = \sqrt{b} + \frac{x}{y}$$

$$a = (b + \frac{x^2}{y^2} + 2\sqrt{b}\frac{x}{y})(\sqrt{b} + \frac{x}{y})$$

$$a = b\sqrt{b} + b\frac{x}{y} + \sqrt{b}\frac{x^2}{y^2} + \frac{x^3}{y^3} + 2b\frac{x}{y} + 2\sqrt{b}\frac{x^2}{y^2}$$

$$a = b\sqrt{b} + 3\frac{bx}{y} + 3\sqrt{b}\frac{x^2}{y^2}$$

$$a - 3\frac{bx}{y} = \sqrt{b}(b + 3\frac{x^2}{y^2})$$

$$\sqrt{b} = \frac{a - 3\frac{bx}{y}}{b + 3\frac{x^2}{y^2}}$$

provided $\sqrt[3]{a} \neq \sqrt{b}$

~~if $\sqrt[3]{a} - \sqrt{b}$ is irrational~~ if $\sqrt{b}$ is irrational $\sqrt[3]{a} + \sqrt{b}$ is irrational. This should similarly uphold if you have $\sqrt[3]{a} + \sqrt{b}$, ~~provided~~ also

$$\sqrt[3]{a} = \sqrt{b} + \frac{x}{y} \rightarrow \sqrt[3]{a} - \frac{x}{y} = \sqrt{b} \rightarrow b = \sqrt[3]{a^2} - 2\sqrt[3]{a}\frac{x}{y} + \frac{x^2}{y^2}$$

$$b - \frac{x^2}{y^2} = \sqrt[3]{a^2} - 2\sqrt[3]{a} \rightarrow (\sqrt[3]{a^4} + 4\sqrt[3]{a^2} - 4a)(\sqrt[3]{a^2} - 2\sqrt[3]{a}) = (b - \frac{x^2}{y^2})^3$$

$(\sqrt[3]{a^4} + 4\sqrt[3]{a^2} - 4a)(\sqrt[3]{a^2} - 2\sqrt[3]{a}) = (b - \frac{x^2}{y^2})^3$ pg 26

$\sqrt[3]{a^6} - 2\sqrt[3]{a^5} + 4\sqrt[3]{a^4} - 8a \cdots - 4a\sqrt[3]{a^2} + 8a^3\sqrt[3]{a} = (b - \frac{x^2}{y^2})^3$

a^2

$\sqrt[3]{a} + \sqrt{5} = \sqrt[6]{a^2} + \sqrt[6]{5^3}$

Not if:

$a = 3, 3 \quad b = 3$

$\sqrt[3]{4} + \sqrt{3} = \sqrt[6]{81}_{3,3,3,3} + \sqrt[6]{27}_{3,3,3}$

$x \in \mathbb{Z}$

$\sqrt[6]{x} \Rightarrow y \in \mathbb{Z}$

$\sqrt[6]{3^4} \therefore \sqrt[6]{3^2}$

$\sqrt[6]{3^4} + \sqrt[6]{5^2} = \sqrt[6]{3^2}\sqrt[6]{3^2} + \sqrt[6]{5^3}$

$\sqrt[3]{a} + \sqrt{5} = \sqrt{5} + \sqrt{a^{\frac{2}{3}}}$

Now that it's a square-root summation previous trick applies, so $a^{\frac{2}{3}}$ must be a square number, so, a must be a cube and b must be a square, for the $\sqrt[3]{a} + \sqrt{5}$ to be rational. The same logic $a^{\frac{2}{3}} = n^2$ $a = n^3$ can be used for subtraction except $\sqrt[3]{a} \neq \sqrt{5}$

Let's say we have $\sqrt[3]{a}+\sqrt[3]{b}+\sqrt[3]{c}$ ny 27

$$\sqrt{a^{\frac{2}{3}}}+\sqrt{b^{\frac{2}{3}}}+\sqrt{c^{\frac{2}{3}}}$$

a, b, and c must be cubes since $a^{\frac{2}{3}}$ must be a square number, and that goes for b and c as well. This idea works for any power, meaning that you can just transform that power into a square root and set it equal to n^2

ex) $\sqrt[75]{a}+\sqrt[32]{b}=$

$$\sqrt{a^{\frac{2}{75}}}+\sqrt{b^{\frac{1}{16}}}$$

$a^{\frac{2}{75}}=n^2 \Rightarrow a=n^{75}\ n\in\mathbb{Z}$

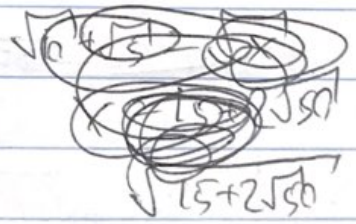
$b^{\frac{1}{16}}=n^2 \Rightarrow b=n^{32}\ n\in\mathbb{Z}$

$\sqrt{6}+\sqrt{30}+\sqrt{15}+\sqrt{3}$

$\sqrt{2}\sqrt{3}+\sqrt{2}\sqrt{3}\sqrt{5}+\sqrt{3}\sqrt{5}+\sqrt{3}$

$(\sqrt{2}+\sqrt{10}+\sqrt{5})(\sqrt{3})$

irrational, irrational



$\sqrt{x}=\sqrt{2}+\sqrt{10}+\sqrt{5}$

$x=(\sqrt{2}+\sqrt{10}+\sqrt{5})(\sqrt{2}+\sqrt{10}+\sqrt{5})$

$x=(17+2\sqrt{20}+2\sqrt{50}+2\sqrt{10})$

$x=(17+4\sqrt{5}+10\sqrt{2}+2\sqrt{10})$ irrational

if x^2 is irrational, x is irrational

assume x is rational and can be expressed in simplest form as $\frac{a}{b}$. $x^2=\frac{a^2}{b^2}$, however x^2 is irrational so this can't be true, hence x is irrational

* it's arguable that the solution only allows 9 terms (3 terms each comprised of 3 terms that can be summed) However what if you take the inner terms of one of the larger terms and combine those for try to get less terms

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Then the problem of proving $\sqrt{a}+\sqrt{b}+\sqrt{c}+\sqrt{d}\ldots$ where at least one number in the square root is a non-square leading to the sum being irrational can be solved by grouping terms and making them 1 square root term as many times as necessary until only 3 terms are left, and the case of 3 terms has already been proven. This also extends to all other roots, as they can be expressed as square roots

ex) $\sqrt{6}+\sqrt{30}+\sqrt{15}+\sqrt{3}$

$\sqrt{x}=\sqrt{6}+\sqrt{30}$ $\quad$ $\sqrt{y}=\sqrt{15}+\sqrt{3}$

'irrational' $x=36+2\sqrt{180}$ $\quad$ $y=18+2\sqrt{45}$

must be irrational due to root on last page $\sqrt{x}=\sqrt{36+2\sqrt{180}}$ $\quad$ $\sqrt{y}=\sqrt{18+2\sqrt{45}}$

If I worked 1 term!

$\sqrt{2}=\sqrt{x}+\sqrt{y}$

$2=(\sqrt{36+2\sqrt{180}}+\sqrt{18+2\sqrt{15}})(\sqrt{36+2\sqrt{180}}+\sqrt{18+2\sqrt{15}})$

$2=36+2\sqrt{180}+18+2\sqrt{15}+2\sqrt{36+2\sqrt{180}}\sqrt{18+2\sqrt{15}}$

$2=36+2\sqrt{180}+18+2\sqrt{15}+2\sqrt{36\cdot18+72\sqrt{15}+36\sqrt{180}+4\sqrt{180\cdot15}}$

$2=$ summation of 3 irrational roots

summation of 3 irrationals must be irrational

roots

if x^2 is irrational, x must be irrational

→ an integer which is not a square (for fractional powers like $\frac{1}{3}$ convert the number to a square root first)

So, the sum of any amount of rational numbers raised to fractional powers is irrational provided at least one of the numbers product of their numerator and denominator, ~~in most simplified form, is a non-square integer.~~ If all ~~numbers~~ rational numbers, when expressed in most simplified form (meaning x and y are integers if the rational number is $\frac{x}{y}$) have their numerator and denominator multiply to a square number (for fractional powers like $\frac{1}{3}$, once again, convert the number to a square root first

2.

a) Figure out how A's magic trick works, and write up a clear, mathematical explanation of how to perform it

Let $x = n$ $n\varepsilon$

There are 4 cases: Rnd down 1st and Rnd down 2nd, Rnd down 1st and don't Rnd down 2nd, Don't Rnd down 1st and Rnd down 2nd, Rnd down 1st and 2nd

multiples of 4 never round down

multiples of 2 but not of 4 round down only on the 2nd time

multiples of 4 +1 round down twice

multiples of 4 -1 round down only on the 1st time

→ if $\frac{x}{4} \in \mathbb{Z}$ then $9 \cdot \frac{x}{4} \in \mathbb{Z}$

→ if $\frac{x}{2} \in \mathbb{Z}$, $\frac{x}{4}$ is not $\in \mathbb{Z}$ then $3 \cdot \frac{x}{2} \in \mathbb{Z}$ and $9 \cdot \frac{x}{4}$ is not

This is because multiplying an integer by an integer yields an integer, and multiplying an integer by a fraction that does not share any factors with the denominator yields a fraction

→ this can be expressed as $\frac{x}{4}+1$, $\frac{x}{4} \in \mathbb{Z}$

1st step $\frac{3(x+1)}{2} = \frac{1}{2}(3x+3) = \frac{3}{2}x + \frac{3}{2}$

$\frac{3}{2}x$ will be an even integer since $\frac{x}{4} \in \mathbb{Z}$

$\frac{3}{2}x + \frac{3}{2}$ will therefore be a fraction, and must be rounded down to $\frac{3}{2}x + 1$

2nd step: $\frac{1}{2}(3)(\frac{3}{2}x+1) = \frac{1}{2}(\frac{9}{2}x+3) = \frac{9}{4}x+\frac{3}{2}$

$\frac{9}{4}x$ will be an integer, since $\frac{x}{4} \in \mathbb{Z}$, therefore $\frac{9}{4}x+\frac{3}{2}$ will have to be rounded down since $\frac{3}{2}$ is a fraction

numbers in the form $x+1$ where $\frac{x}{4} \in \mathbb{Z}$ will have to be rounded down twice

multiples of 4 -1:

$x-1$, $\frac{x}{4} \in \mathbb{Z}$

$\frac{1}{2}(3)(x-1) = \frac{1}{2}(3x-3) = \frac{3}{2}x - \frac{3}{2}$

$\frac{3}{2}x$ is an integer, because $\frac{x}{4} \in \mathbb{Z}$

$\frac{3}{2}x - \frac{3}{2}$ therefore is not an integer, and will have to be rounded down to $\frac{3}{2}x-2$

$\frac{1}{2}(3)(\frac{3}{2}x-2) = \frac{1}{2}(\frac{9}{2}x-6) = \frac{9}{4}x-3$

$\frac{9}{4}x$ is an integer, because $\frac{x}{4} \in \mathbb{Z}$, therefore $\frac{9}{4}x-3$ is an integer, because an integer $-$ an integer is an integer, and so the number will not need to be rounded down

taking quotient $\times 4 + 3$ works for numbers $x-1$ $\frac{x}{4} \in \mathbb{Z}$

taking quotient $\times 4 + 1$ works for numbers $x+1$ $\frac{x}{4} \in \mathbb{Z}$

B4

One way to solve this problem seems to be considering the multiple of 4, and then adding numbers based on how far off from the last multiple of 4 the number is

[redacted]

Since multiply by 3 twice and dividing by 9 cancels out, the only part of the trick that isn't multiplying your number by 4 is the rounding down part.

[redacted]

take a number like 5, where you round down twice

take a number

So just multiply the quotient by 4 and use the amount of times partner B rounded down to figure out what you needed to add to that total

Case 2: no rounding down

$k = 4q$ k = original number, q = quotient

Case 2: rounding down happened on the first time but not the second

$k = 4q + 3$

Case 3: no rounding down happened on the first time, but it happened on the second time

$k = 4q + 2$

Case 4: rounding down occured both times

$k = 4q + 1$

Explanation for each case

Case 1:

X = original numbe $\frac{X}{4} \in \mathbb{Z}$ (no rounding)
q = quotient

$\frac{qx}{4} \div q = q$

$\frac{x}{4} = q$

$x = 4q$

Case 2: $x+3$ = original number, $\frac{x}{4} \in \mathbb{Z}$ pg 6

$3\left(\frac{3x+9}{2} - \frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{9}\right) = q$ $\quad q$ = quotient

$3\left(\frac{3}{2}x + 4\right)\left(\frac{1}{2}\right)\left(\frac{1}{9}\right) = q$

$\left(\frac{9}{2}x + 12\right)\left(\frac{1}{2}\right)\left(\frac{1}{9}\right) = q$

$\frac{1}{9}\left(\frac{9}{4}x + 6\right) = q$

$\frac{1}{4}x + \frac{2}{3} = q$

$\frac{1}{4}x = q - \frac{2}{3}$

$x = 4q - \frac{8}{3}$

$x + 3 = 4q + \frac{1}{3}$ → ~~this is not what we want~~

The quotient value is not exact so lets focus on the value before dividing by 9

~~$3\left(\frac{3x+9}{2} - \frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{9}\right) = q$~~

$3\left(\frac{3x+9}{2} - \frac{1}{2}\right)\left(\frac{1}{2}\right)$

$3\left(\frac{3}{2}x + 4\right)\left(\frac{1}{2}\right)$

$\frac{9}{4}x + 6 = R$

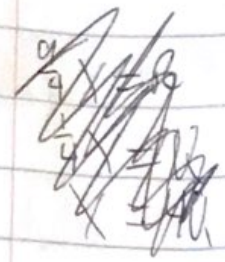
~~$\frac{4}{9}\left(x + 2\frac{2}{3}\right) = \frac{4}{9}R$~~

$\frac{9}{4}\left(x + 2\frac{2}{3}\right) = R$ → this will be a decimal when divided by 9, so remove it now, but add it back later

$\frac{9}{4} \cdot \frac{1}{9} \cdot x$ will be integer

$\left(\frac{9}{4} \cdot \frac{1}{9} = \frac{9}{36} < 1\right)$

R = value of equations before dividing by 9



$3\left(\frac{3x+9}{2}-\frac{1}{2}\right)\left(\frac{1}{4}\right)\left(\frac{1}{9}\right)=q$

$3\left(\frac{3}{2}x+4\right)\left(\frac{1}{2}\right)\left(\frac{1}{9}\right)=q$

$\left(\frac{9}{4}x+6\right)\frac{1}{9}=q$

$\frac{1}{4}x=q-\frac{6}{9}$

$x=4q-\frac{8}{3}$

$x+3=4q+\frac{1}{3}$

now, since the quotient does not include the remainder, add back the $2\frac{2}{3}$ (previous step)

$x+3=4q+\frac{1}{3}+2\frac{2}{3}$

$x+3=4q+3$

Case 3: let $x+2$ = original number $\frac{x}{4}\in\mathbb{Z}$

q = quotient

R = number before divide by 4

$\left[3\left(\frac{3x+6}{2}\right)\left(\frac{1}{2}\right)-\frac{1}{2}\right]\left(\frac{1}{9}\right)=q$

$\left[\left(\frac{9x+18}{2}\right)\left(\frac{1}{2}\right)-\frac{1}{2}\right]\left(\frac{1}{9}\right)=q$

$\left(\frac{9x+18}{4}-\frac{1}{2}\right)\left(\frac{1}{9}\right)=q$

$\left(\frac{9x}{4}+4\right)\left(\frac{1}{9}\right)=q$

$3\left(\frac{3x+6}{2}\right)\left(\frac{1}{2}\right)-\frac{1}{2}=R$

$\left(\frac{9x}{2}+9\right)\left(\frac{1}{2}\right)-\frac{1}{2}=R$

$\frac{9x}{4}+4=R$

$\frac{9}{4}\left(x+\frac{16}{9}\right)=R$

↳ $\frac{16}{9}\cdot\frac{1}{9}=\frac{16}{91}<1$ Can't show up in quotient

$\left(\frac{9x}{4}+4\right)\left(\frac{1}{9}\right)=q$

$\frac{x}{4}+\frac{4}{9}=q$

$\frac{x}{4}=q-\frac{4}{9}$

$X=4q-\frac{16}{9}$

$x+2=4q+\frac{2}{9}$

add back the $\frac{16}{9}$ that didn't end up as part of the quotient

$x+2=4q+2$

Case 4:

let $x+1$ = original number, $\frac{x}{4}\in\mathbb{Z}$

q = quotient R = number before dividing by 9

$3\left(\frac{3(x+1)}{2}-\frac{1}{2}\right)\left(\frac{1}{2}\right)-\frac{1}{2}=R$

$3\left(\frac{3x+3}{2}-\frac{1}{2}\right)\left(\frac{1}{2}\right)-\frac{1}{2}=R$

$3\left(\frac{3x}{2}+1\right)\left(\frac{1}{2}\right)-\frac{1}{2}=R$

$\left(\frac{9x}{2}+3\right)\left(\frac{1}{2}\right)-\frac{1}{2}=R$

$\frac{9x}{4}+\frac{3}{2}-\frac{1}{2}=R$

$\frac{9x}{4}+1=R$

$\frac{9}{4}\left(x+\frac{4}{9}\right)=R$

↳ $\frac{4}{9}\cdot\frac{1}{4}\notin\mathbb{Z}$, so we will need to add back $\frac{4}{9}$ at the end

$$\left[3\left(\frac{3(x+1)}{2}-\frac{1}{2}\right)\left(\frac{1}{2}\right)-\frac{1}{2}\right]\left(\frac{1}{9}\right)=q$$

$$\left[3\left(\frac{3x}{2}+1\right)\left(\frac{1}{2}\right)-\frac{1}{2}\right]\left(\frac{1}{9}\right)=q$$

$$\left[\left(\frac{9x}{2}+3\right)\left(\frac{1}{2}\right)-\frac{1}{2}\right]\left(\frac{1}{9}\right)=q$$

$$\left(\frac{9x}{4}+1\right)\left(\frac{1}{9}\right)=q$$

$$\frac{x}{4}+\frac{1}{9}=q$$

$$\frac{x}{4}=q-\frac{1}{9}$$

$$x=9q-\frac{4}{9}$$

$$x+1=9q+\frac{5}{9}$$

$$x+1=9q+1$$

b) What variants of this trick can you come up with using the same principles? Can you change the numbers 2, 3, and 9 in the trick or maybe the operations involved? What about the information you ask for?

It's convenient to have the dividing number, in this case 9, equal to the square of the multiplier, in this case 3, because then, besides for the rounding down, they cancel out.

I will try making those numbers 5 and 25 not 10 (not 4 and 16 because then the rounding down part would be irrelevant making the trick less complex and easier to figure out

Test numbers: 16, 17, 18, 19, 20 (does it repeat in any sense over a pattern of 4, which it probably will

16 → 4 17 → 4 (4.2 rounded down on the first time)
18 → 4 (4.48, rounded down on the 2nd time)
19 → 4 (4.68, rounded down both times)
20 → 5

The idea of each group of 4 yielding the same quotient, and the next group yielding a quotient 1 higher seems independent of the 3 and 9, provide the former squared equals the latter

However, now numbers in the form of $x+3$ $\frac{x}{4} \in \mathbb{Z}$ get rounded down twice and numbers in the form $x+1$ $\frac{x}{4} \in \mathbb{Z}$ are only rounded down on the first time

$5\left(\frac{5}{2}x+\frac{5}{2}-\frac{1}{2}\right)\left(\frac{1}{2}\right) \rightarrow \left(\frac{25x}{2}+10\right)\left(\frac{1}{2}\right) \rightarrow \frac{25x}{4}+5 \rightarrow$ integer

$3\left(\frac{3x}{2}+\frac{3}{2}-\frac{1}{2}\right)\left(\frac{1}{2}\right) \rightarrow \left(\frac{9x}{2}+3\right)\left(\frac{1}{2}\right)$

3-1=2 5-1=4

$\frac{1}{2}$ is always subtracted, but numbers like 5, when subtracted by 1 produce multiples of 4, as will every other odd number (3 no, 5 yes, 7 no, 9 yes etc.)

With that in mind, the magic trick could've gone like this:

A: Want to see a magic trick?
B: Sure, how does it go?
A: Think of any nonnegative integer
B: Okay
A: Multiply it by an odd integer N
B: okay...
A: Now divide it by 2, and if you get a decimal round down. Let me know if you had to round down.
B: I had to round down
A: Multiply by that same odd integer N. Then divide by 2 and say if you had to round down
B: I did not have to round down
A: Great, now how many times does the odd integer squared, N^2 go into your number?
B: The quotient is 4
A: Can the odd integer N be expressed as 1 + a multiple of 4?
B: No
A: Your original number is 19

If the student had answered yes to the final question, the original number would have been 17

Changing 2:

What if the 2 becomes a 5 (if it's a 3 it will cancel out with the multiplier)

I think there will be overlapping occurrences of amount of times the number is rounded down, and so performing the x3/5 twice won't be enough. Each quotient has 25 possibilities, based off of what happened with 2

~~4, which is almost enough to distinguish between the 25 possibilities~~

$2^5 = 32$ $2^4 = 16$, so for 25 combinations you should do the process 5 times. It is also possible 4 times wouldn't work because of similarity of answers (which would be a problem even if there were enough combinations to allow 25 distinct results

Using 3 would make the example simpler, and pg 13
this can be done by changing the multiplier to
5

There is still a problem, as both a result ending
in 0,33... and 0,66... would make partner B
answer "yes" to the question of whether or not they
rounded down. More information is needed, so studat
A should also ask if they had to subtract a
value of greater than 0,5.

The 9 cases:

Case 1: x = original number, $\frac{x}{9} \in \mathbb{Z}$
Case 2: $x+1$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 3: $x+2$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 4: $x+3$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 5: $x+4$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 6: $x+5$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 7: $x+6$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 8: $x+7$ = original number, $\frac{x}{9} \notin \mathbb{Z}$
Case 9: $x+8$ = original number, $\frac{x}{9} \notin \mathbb{Z}$

Case 1:

$5\left(\frac{5x}{3}\right)\left(\frac{1}{3}\right) = \frac{25x}{9}\left(\frac{1}{25}\right) = \frac{x}{9} = q \quad x = 9q \quad q = \text{quotient}$

Case 2:

$\frac{1}{25}\left(\left\{5\left[\frac{1}{3}(5x+5)-\frac{2}{3}\right]\right\}\left(\frac{1}{3}\right)-\frac{2}{3}\right)=q$

$\frac{1}{25}\left[5\left(\frac{5}{3}x+1\right)\right]\frac{1}{3}-\frac{2}{3}\}=q$

$\frac{1}{25}\left[\left(\frac{25}{3}x+5\right)\frac{1}{3}-\frac{2}{3}\right]=q$

$\frac{1}{25}\left(\frac{25}{9}x+1\right)=q$

~~$25x+1=q$~~

$\frac{1}{9}x+\frac{1}{25}=q$

$\frac{1}{9}x=q-\frac{1}{25}$

$x=9q-\frac{9}{25}$

$x+1=9q+\frac{16}{25}$

add back $\frac{9}{25}$

$x+1=9q+1$

$\frac{25}{9}x+1=R$ R = number before divide by 25

$\frac{25}{9}\left(x+\frac{1}{25}\right)=R$

→ won't show up in equation

~~Another way of looking at the above solution is that since we found that $x=9q$~~

→ this format is true regardless if it's $x+1$ or $x+5$ the term you add to x, is the same as the term you add to $9q$ ($x=9q$)

However, the question for each case is the decimals you have. For the prior example, I just used an example on my calculator, but will now algebraically do it

First disregard the x term, because it will always be an integer

$\frac{5(1)}{3} = \frac{5}{3} \Rightarrow \frac{5}{3} - 1 = \frac{2}{3}$ 1st round down will be by $\frac{2}{3}$

$\frac{5}{3} - \frac{2}{3} = 1 \rightarrow \frac{5(1)}{3} = \frac{5}{3}$ $\frac{5}{3} - 1 = \frac{2}{3}$ 2nd round down by $\frac{2}{3}$

Case 3: $\frac{5(2)}{3} = \frac{10}{3} =$ [illegible] $3\frac{1}{3} - 3 = \frac{1}{3}$

$3\frac{1}{3} - \frac{1}{3} = 3 \rightarrow \frac{5(3)}{3} = 5$ no remainder

Case 4: $\frac{5(3)}{3} = 5$ no remainder

$5 - 0 = 5 \rightarrow \frac{5(5)}{3} = 8 + \frac{1}{3}$

Case 5: $\frac{5(4)}{3} = \frac{20}{3} = 6\frac{2}{3} - 6 = \frac{2}{3}$

$6\frac{2}{3} - \frac{2}{3} = 6 \rightarrow \frac{5(6)2}{3} = 10$ no remainder

Case 6: $\frac{5(5)}{3} = \frac{25}{3} = 8\frac{1}{3} - 8 = \frac{1}{3}$

$8\frac{1}{3} - \frac{1}{3} = 8$ $\frac{8(5)}{3} = \frac{40}{3} = 13\frac{1}{3} - 13 = \frac{1}{3}$

Case 7: $\frac{5(6)2}{3} = 10$ no remainder

$\frac{5 \cdot 10}{3} = \frac{50}{3} = 16\frac{2}{3} - 16 = \frac{2}{3}$

Case 8: $\frac{5(7)}{3} = \frac{35}{3} \Rightarrow 11\frac{2}{3} - 11 = \frac{2}{3}$

$11\frac{2}{3} - \frac{2}{3} = 11$ $\frac{5(11)}{3} = \frac{55}{3} =$ [illegible] $18\frac{1}{3} \rightarrow \frac{1}{3}$

Case 9: $\frac{5(8)}{3} = \frac{40}{3} = 13\frac{1}{3} \rightarrow 13\frac{1}{3} - 13 = \frac{1}{3}$

$13\frac{1}{3} - \frac{1}{3} = 13 \rightarrow \frac{5(13)}{3} = \frac{65}{3} = 21\frac{2}{3} \rightarrow \frac{2}{3}$

Reference Chart

Original number	1st remainder	2nd remainder
X	none	none
X+1	$\frac{2}{3}$	$\frac{2}{3}$
X+2	$\frac{1}{3}$	none
X+3	none	$\frac{1}{3}$
X+4	$\frac{2}{3}$	none
X+5	$\frac{1}{3}$	$\frac{1}{3}$
X+6	none	$\frac{2}{3}$
X+7	$\frac{2}{3}$	$\frac{1}{3}$
X+8	$\frac{1}{3}$	$\frac{2}{3}$

These values will of course change if the 3 changes, but also if the 5 is changed. ~~The pattern for the 3 does~~ ~~rotates between a given number~~

Forexample take the number 4. When 5 is the multiplier you get a remainder (the first one) of 2, yet for 7 it's 1.

I used odds but something like 8, 9, 10 would also have the pattern (in this case 2, 0, 2 for 4), and of course the full pattern is on a cycle of 9

For 9, you get 0 remainders, and the cycle repeats. pg 17
The cycle spans every 3 numbers, because every multiple of 3 just adds to the whole number groups of 3 and leaves the remainder unaffected.

The process for coming up with a divider other than 3 is just like the one shown previously with figuring out the remainder in each case, however as the numbers get bigger you will have to ask for more and more information about the value of the remainder (instead of asking if its greater than 0.5 you may have to provide more ranges. Alternatively you can just ask for the actual value of the remainder)

Variation of trick with multiplier of 5 and divider of 3:

A: Hey, want to see a magic trick?
B: Sure, how does it go?
A: Think of any nonnegative integer
B: okay
A: Multiply it by 5
B: okay
A: Now divide it by 3, and if you get a decimal round down. Let me know if you round down, and tell me if by rounding down you subtracted a value from your number that is greater than $\frac{1}{2}$

B: I rounded down, and subtracted a value of greater than 0.5
A: Multiply by 5 and divide by 3 again. Tell me again if you had to round down again, and if you subtracted a value of greater than $\frac{1}{2}$ again
B: I rounded down, and subtracted a value of greater than $\frac{1}{2}$
A: Now divide by 25 and tell me what the quotient is
B: The quotient is 2
A: So your original number was 19

What happens if we introduce negative numbers. Then rounding down would be the opposite, for example if you had -18.333... it would round down to -19. However, in part of the trick you could just say to round such that you are left with a smaller absolute value, or figure out a new chart of corresponding roundings to numbers for negatives. ~~Finally you can add a step of squaring the quotient so~~ Finally, you can use the quotient, to see if the original number was negative or positive (quotient will match sign of original number unless the quotient is 0, in which case you could just ask if the number is negative). Here is an example of the original magic trick that includes negative numbers:

A: Hey, wanna see a magic trick?
B: Sure, how does it go?
A: Think of any integer
B: Okay
A: Multiply it by 3
B: Okay
A: Divide it by 2, and if you get a decimal round such that your resulting value has a lower absolute value. Tell me if you need to round down.
B: ~~Multiply by 3 and divide by 2~~ I had to round
A: Multiply by 3 and divide by 2 again. Let me know if you round down
B: I did not have to round down again
A: Now divide by 9 and tell me the quotient
B: The quotient is -4
A: Your original number was -19

What if instead of multiplying by 3, you add 32. Then there would be no need to divide by 9, so that part of the trick would not be necessary.

Then it's like you're multiplying your number by 1, which is 1 greater than a multiple of 4, so the pattern of remainders from before when I was multiplying by 5 and dividing by 2 should be the same as the pattern for this variation of the trick.

~~Case 1:~~

However, since we're adding 3 there might be some translation to the pattern assuming the pattern remains

Case 1: ~~$\frac{x}{4} \in \mathbb{Z}$~~ x=original number, $\frac{x}{4} \in \mathbb{Z}$
Case 2: $x+1$=original number, $\frac{x}{4} \in \mathbb{Z}$
Case 3: $x+2$=original number, $\frac{x}{4} \in \mathbb{Z}$
Case 4: $x+3$=original number, $\frac{x}{4} \in \mathbb{Z}$

Case 2:



$\frac{1}{2}(x+3) \rightarrow \frac{1}{2}x + \frac{3}{2} \rightarrow \frac{3}{2} - 1 = \frac{1}{2}$
$\frac{3}{2} - \frac{1}{2} = 1$ $\frac{1}{2}(1+3) = 2$ none

Case 2:

$\frac{1}{2}(x+1+3) \rightarrow \frac{1}{2}x + 2$ none
$\frac{1}{2}(2+3) = 5(\frac{1}{2}) = \frac{5}{2}$ $2\frac{1}{2} = 2$ ~~$\frac{1}{2}$~~

Case 3:
$\frac{1}{2}(x+2+3) \rightarrow \frac{1}{2}x + \frac{5}{2} \rightarrow 2\frac{1}{2}$ $\frac{1}{2}$
$2\frac{1}{2} - \frac{1}{2} = 2$ $\frac{1}{2}(2+3) = \frac{5}{2} \rightarrow 2\frac{1}{2}$ $\frac{1}{2}$
Case 4: $\frac{1}{2}(x+3+3) = \frac{1}{2}(x+6) \rightarrow \frac{1}{2}x + 3$ none
$\frac{1}{2}(3+3) = \frac{1}{2}(6) = 3$ none

The way the cases worked when the multiplier was pg 21
5, one more than a square was as follows

X; none, none
X+1; rounded, none
X+2; none, rounded
X+3; rounded, rounded

These are the cases when 3 is added twice

X; rounded, none
X+1; none, rounded
X+2; rounded, rounded
X+3; none, none

X is translated 1 ahead of the regular pattern because of the addition of 3 shifting the multiples that go into X

Adding 0 obviously shifts the pattern by 0, and adding 3 shifts the pattern forward one. (or forward 3) Adding 1 then likely shifts the pattern back 1, and 2 shifts it forward 2. If that's true, then when I add one to 8 I should have to round twice. If I add two to 8 it should have to round the 2nd time

(8+1)=9 → 9/2=4.5 → (rnd) → 4 → 4+1=5 → 5/2=2.5 → (rnd) → 2

(8+2)=10 → 10/2=5 → (no rounding) → 5+2=7 → 7/2=3.5 → (rnd) → 3

* For addition negatives would be more complex as they could turn positive

Using this information, a new magic trick can be created

A: Wanna see a magic trick?
B: Sure, how does it go?
A: Think of a number A non-negative integer
B: Okay
A: Add another non-negative integer, N, and tell me what N is equal to
B: okay, N=5
A: Now divide by 2, and if you get a decimal round down Tell me if you rounded do
B: I did not have to round down
A: Add N and divide by 2 again and tell me if you had to round down
B: I had to round down
A: What is the value of your number now?
B: 8
A: Your original number was [redacted]

$\frac{1}{2}\left[\frac{1}{2}(X+5)+5\right]=8$

$\frac{1}{2}\left[\frac{1}{2}X+\frac{15}{2}\right)=8$

$\frac{1}{4}X+\frac{15}{4}=8$

$\frac{1}{4}X=\frac{17}{4}$

$X=17$

16 is nearest power of four less than 17

~~16 is nearest~~

N=5 and 5 is left a power of 4, Adding 1 shifts the pattern 3 forward (would have said 1 back if we used nearest larger power to 17), 16+3=19

Alternatively you could just solve purely in equations style pg 23 because all the information is known.

$\frac{1}{2}\left[\frac{1}{2}(x+5)+5\right]-\frac{1}{2}=8$

$\frac{1}{2}\left(\frac{1}{2}x+\frac{15}{2}\right)=\frac{17}{2}$

$\frac{1}{2}x+\frac{15}{2}=17$

$\frac{1}{2}x=\frac{19}{2}$

$x=19$

Another way to change the trick would be to ask for different information.

Instead of asking for the ~~[illegible]~~ amount of times 9 goes into the number, you can ask for the value of their number divided by 9 (including decimals). Then you wouldn't have to ask if they rounded down.
Example:

A: Want to see a magic trick?
B: Sure, how does it go?
A: Think of an integer
B: Okay
A: Multiply it by 3
B: Okay

nevermind, this doesn't work because you don't know if they rounded down, so you don't know difference between quotient x4 and their original number

A: Now divide it by 2, and if you get a decimal round down
B: Okay
A: Now multiply by 3 and divide by 2 again, and round down if necessary again
B: Okay
A: Now divide by 9 and tell me the result
B:

You could ask if they rounded down and ask for their quotient, but then let them choose any numbers, although then they'd have to say how much they rounded down by.

A: Want to see a magic trick?
B: Sure
A: Think of any number
B: okay
A: Multiply it by a real number M, and tell me the value of M
B: okay, M=7
A: Now divide it by a real number B, and tell me B's value. If you get a decimal, round down, and tell me what value C you rounded down by
B: Okay, $B=\frac{5}{8}$ and $C=0.8$
A: Multiply by a real number A and divide by a real number D and tell me the values of A and D. Then round down and tell me what value F you rounded down by.

Can you type in a negative number?

a if their total is zero, ask if the original number was negative or positive

B: okay, $A=\frac{-4}{3}$ $D=\frac{7}{9}$ and $F=\frac{3}{7}$ pg 25
A: Great, now divide your number by Z and tell me Z's value as well as your current total
B: Z=5, and my current total is -72.6
A: your original number was 19

For this one, just use the equation

$\{[A(\frac{nx}{B}-C)](\frac{1}{D})-F3(\frac{1}{Z})=T$ T=current total

C and F should be subtracted if positive and added if negative
You can determine their signs by working backwards for example, T is negative and Z is positive, so $[A(\frac{nx}{B}-C)](\frac{1}{D})-F$ must be negative making F negative. ~~A was negative making~~ D was positive meaning $A(\frac{nx}{B}-C)$ is negative. A is negative meaning $\frac{nx}{B}-C$ is positive so C is positive (subtracting C will never change the sign

Another way of switching the information is asking how far off the nearest smaller multiple of 4 is instead of asking if you rounded down

- if quotient was 0 ask if number was negative or not

A: Hey wanna see a magic trick
B: Sure
A: Think of an integer
B: okay
A: Multiply it by 3
B: Okay
A: Now divide it by 2 and round such that your absolute value decreases
B: okay
A: Multiply by 2 and divide by 2 again, and round again if necessary
B: okay
A: Now divide by 4 and tell me the quotient. Tell me how far your number is from the nearest lower multiple of 4 as well
B: The quotient is 4 and the distance to the nearest lower multiple of 4 is 3
A: your number is 19

in terms of absolute value

$x = 4q + R$ q = quotient R = distance to nearest lower multiple of 4

You could also change some numbers, for example you could change the 2 being the dividing number, if the dividing number was N, $x = Nq + R$, R = distance to nearest multiple of N^2 (in absolute value)

you could also change the 3, but it would make sense to keep the first dividing number, originally 4, as equal to the multiplier squared

the multiplier and final dividing number should stay the same, so they cancel out and you are left with $\frac{x}{n^2} \cdot n^2$, which gives you x. Then the decimals are made up for by giving the distance to the nearest multiple (which, in some sense, is what ~~the~~ decimal is) and you can get the original number

3. It's hard to keep up with the fashionable trends of the day, and mathematical fashion is no exception. The fashion season of 2024 has the following unspoken fashion rules: /1

- 1 is fashionable.
- If a and b are fashionable, then so is $\frac{1}{a+b}$.
- All other numbers are "so last year" and thus not fashionable.

a) Find the smallest and largest fashionable numbers, and prove that they are the smallest and largest.

Logically the smallest number must be $\frac{1}{2}$ and the largest number must be 1.

This is because the smallest number will be created when $a+b$ is as big as possible. The first number generated by the sequence $\frac{1}{a+b}$, where 1 is the 1st number in the set is $\frac{1}{2}$. $\frac{1}{2}$ can only be added to itself (which generates 1) and 1, which also generates a fraction between $\frac{1}{2}$ and 1 ($\frac{2}{3}$). Since any fraction between $\frac{1}{2}$ and $1+\frac{1}{2}$ or $+1$ generates a fraction between $\frac{1}{2}$ and 1

(This is true since $\frac{1}{\frac{1}{2}+1} = \frac{2}{3}$, $\frac{1}{\frac{1}{2}+\frac{1}{2}} = 1$, $\frac{1}{1+1} = \frac{1}{2}$ and $\frac{1}{1+\frac{1}{2}} = \frac{2}{3}$, and it the value of those numbers are between $\frac{1}{2}$ and 1)

~~most~~ any values between those numbers must also be between $\frac{1}{2}$ and 1 (in other words if the highest and lowest current value generate those numbers for ~~that~~ something like $\frac{1}{a+b}$ which is just division, numbers within those values also must generate responses in that range.

For example $\frac{1}{\frac{2}{3}+1} = \frac{3}{5}$ and $\frac{1}{\frac{2}{3}+\frac{1}{2}} = \frac{6}{7}$ because even if we added something as great as 1 or as small as $\frac{1}{2}$ to $\frac{1}{2}$ or 1, we would still be in the range of $\frac{1}{2}$ to 1.

Additionally $\frac{1}{\frac{2}{3}+\frac{2}{3}} = \frac{3}{4}$, and this must be between $\frac{1}{2}$ and 1 because even if we were to add $\frac{1}{3}$ to each term in the denominator (making it 1+1) or subtract $\frac{1}{6}$ to each term in the denominator (making it $\frac{1}{2}+\frac{1}{2}$) we would still be in the range of $\frac{1}{2}$ to 1. This applies to every example of adding two numbers between $\frac{1}{2}$ and 1, because they by definition are less than 1 (so you can add n to one term and b to the other to get 1 where ~~n and b are greater than~~ n and b are positive) and

greater than $\frac{1}{2}$.

~~For the nth case the~~

However, while I believe this logic is correct, it is not a rigorous proof. The logic I used did seem similar to induction, as it involves proving a specific case of the first number in the sequence (P(a)) and logic that involved the fact that the kind of numbers generated by P(a) (or generated by the numbers generated by P(a) or really any stemming from those numbers) won't ever lead to numbers outside the range of $\frac{1}{2}$ to 1, and nor will the numbers generated by those numbers etc. etc.

$\frac{1}{2} \leq a \leq 1$

$\frac{1}{2} \leq b \leq 1$

$\frac{1}{a+b} = C$

Assume $\frac{1}{2} \leq \frac{1}{a+b} \leq 1$, $\frac{1}{a+b} = C$

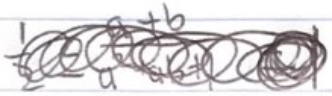
This is a sequence where the new inputs are generated by the outputs of members of the sequence, and since the number that was given as the original was 1, P(a) will be 1.

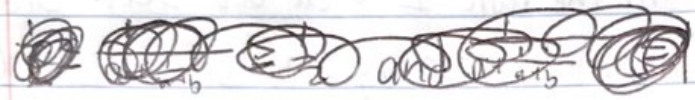
$\frac{1}{2} \leq \frac{1}{1+1} \leq 1$

$\frac{1}{2} \leq \frac{1}{2} \leq 1$ True

$\frac{1}{2} \leq \frac{1}{a+c} \leq 1$

$\frac{1}{2} \leq \frac{1}{a+\frac{1}{a+b}} \leq 1$





$\frac{1}{a+\frac{1}{a+b}} \leq 1$ and $\frac{1}{a+\frac{1}{a+b}} \geq \frac{1}{2}$

$a + \frac{1}{a+b} \geq 1 \quad a + \frac{1}{a+b} \leq 2$

$\frac{a^2+ab+1}{ab} \geq 1 \quad \frac{a^2+ab+1}{ab} \leq 2$

$a^2+ab+1 \geq ab \quad a^2+ab+1 \leq 2a+2b$

$a^2+ab-2a-2b+1 \leq 0$

$a(a-2)+b(a-2)+1 \leq 0$

$(a+b)(a-2)+1 \leq 0$

a and b are both positive and greater than 0, so a^2+ab+1 must be greater than 1. ab is 1 at most since a is 1 at most and b is 1 at most. Therefore $a^2+ab+1 \geq ab$ ($a^2+ab+1 > ab$).

Regarding the 2nd equation, $(a+b)(a-2)+1 \leq 0$, pg 5
$a+b$ must at least be 1, and $a-2$ will have an absolute value of at least 1 and will always be negative, meaning it will at least result in an absolute value of 2 and be negative, meaning when you add one you'll either have a negative left over or nothing, meaning your quantity will be ≤ 0.

This may be clearer if a negative is pulled out:

$-(a+b)(2-a)+1 \leq 0$

$(a+b)(2-a)-1 \geq 0$

$a+b$ will at least be 1, and so will $2-a$, meaning there product will at least be 1, meaning when you subtract 1 you must at least get 0, (≥ 0)

However, this proof used $\frac{1}{a+c}$ where $C = \frac{1}{a+b}$, and not all members of the set will be the sum of numbers one of whom generated the other (a and c). The following proof is for any two numbers that generate a new number

$\frac{1}{2} \leq \frac{1}{a+b} \leq 1$, $\frac{1}{a+b} = C$ Ps 6

$\frac{1}{2} \leq \frac{1}{d+e} \leq 1$, $\frac{1}{d+e} = f$

$\frac{1}{2} \leq \frac{1}{1+1} \leq 1$

$\frac{1}{2} \leq \frac{1}{2} \leq 1$ this works for both C and F

Wait I should just assume $\frac{1}{2} \leq a \leq 1$ and $\frac{1}{2} \leq b \leq 1$, and make $C = \frac{1}{a+b}$, there's no point to the extra steps I was doing

$P(n) \Rightarrow C = \frac{1}{1+1}$, $C = \frac{1}{2}$ $\frac{1}{2} \leq C \leq 1$ True

$\frac{1}{a+b} = C$

$a+b = \frac{1}{C}$

Since $\frac{1}{2} \leq a \leq 1$ and $\frac{1}{2} \leq b \leq 1$, $a+b$ must at least be 1 and at most be 2, meaning C is between $\frac{1}{1}$ and $\frac{1}{2}$, meaning $\frac{1}{2} \leq C \leq 1$

Therefore we can conclude by mathematical induction that this range is valid for every number generated after 1 by this sequence

b) Which numbers are fashionable? Can you characterize the set of all fashionable numbers?

pg 7

Since the prior proof just proved the range of possible numbers being from $\frac{1}{2}$ to 1 without any domain restrictions within that range, I think there is a chance any number within the range is possible.

$\frac{1}{1+\frac{1}{2}} = \frac{2}{3}$ $\quad \frac{1}{\frac{1}{2}+\frac{2}{3}} = \frac{6}{7}$ $\quad \frac{1}{\frac{1}{2}+\frac{6}{7}} = \frac{14}{19}$

no clear pattern and only small numbers encompasses part of possible

diff 2 3 4

Sequence: $1, \frac{1}{2}, \frac{2}{3}, \frac{3}{5}, \frac{6}{7}, \frac{3}{4}, \frac{5}{8}, \frac{10}{11}, \frac{15}{19}, \frac{7}{13}, \frac{14}{19}, \frac{21}{32}, \frac{35}{51}, \frac{7}{12}$

5

denom sum

Denom: 1, 2, 3, 5, 7, 4, 8, 11, 19, 13, 19, 32, 51, 12

Interesting, each denom can be expressed as the sum of previous denoms, which makes sense because the sequence is all about sums of denominators (ofts)

~~This summation rule also applies for they~~

However, the above rule applies more so to the generating of the set rather than its characterization

Let's see if there's any number which I can't make that between $\frac{1}{2}$ and 1:

$\frac{9}{13}$; $\frac{9}{13} = \frac{1}{x}$ $9x = 13$ $x = \frac{13}{9}$ $a+b = \frac{13}{9}$

$\frac{13}{9}$ can be expressed in as many terms as neccessary really, as if a and b add up to $\frac{13}{9}$ but one or both aren't on the list then those fractions can be looked at to determine if they can be expressed as $\frac{1}{a+b}$ using sequence numbers.

$a+b+c+d_{...} = \frac{13}{9}$

$a+b+c+d_{...} = \frac{4}{9}$ → $\frac{9}{4}$ outside domain

not really $a+b+c+d_{...}$ as if $a+b = \frac{13}{9}$ and a is not a sequence number $a = \frac{1}{c+d}$ not $c+d$

$\frac{13}{9} - \frac{2}{3} = \frac{7}{9}$

$\frac{9}{7} = \frac{2}{3} + \frac{13}{21}$

Both are sequence numbers ($\frac{13}{21}$ is shown to be on next page)

Examine sequence numbers generated by 1 (and one other number) pg 9

$1, \frac{1}{2}, \frac{2}{3}, \frac{3}{5}, \frac{5}{8}, \frac{8}{13}, \frac{13}{21}, \frac{21}{34}, \frac{34}{55}$

This sequence is like the Fibonacci sequence, and the reason the fibonacci numbers are generated is that when one is added you find the common denominator, which is just the denominator of the other number, so the numerator gets increased by the value of the denominator and the denominator becomes the numerator. The new numerator is just the denominator from last time, and the new denominator is the value of the original numerator plus the denominator. Additionally the starting numerator and denominator, $\frac{1}{1}$, are both fibonacci numbers, making the sequence akin to the fibonacci numbers (besides for the fact that the fibonacci numbers include 0). (This means any number that is expressed as a fibonacci number over the next fibonacci number not including 0 is a sequence number)

Let's do this again but with a different starting number

Using 1 and $\frac{6}{7}$

$1, \frac{6}{7}, \frac{7}{13}, \frac{13}{20}, \frac{20}{33}, \frac{33}{53},$

Same pattern appears

If a number $\frac{n}{x}$ is a sequence number, $\frac{x}{n+x}$ is a sequence number

This just is using the number one, so more rules like this can be created by adding $\frac{1}{2}$ and adding the number to itself.

$$\frac{1}{\frac{x}{n}+\frac{1}{2}} = \frac{1}{\frac{2x+n}{2n}} = \frac{2n}{2x+n}$$

if $\frac{x}{n}$ is a sequence number $\frac{2n}{2x+n}$ is a sequence number

$$\frac{1}{\frac{x}{n}+\frac{x}{n}} = \frac{1}{\frac{2x}{n}} = \frac{n}{2x}$$

if $\frac{x}{n}$ is a sequence number, $\frac{n}{2x}$ is a sequence number

---

$1, \frac{1}{2}, \frac{2}{3}, \frac{3}{5}, \frac{6}{7}, \frac{3}{4}, \frac{5}{8}, \frac{10}{11}, \frac{15}{19}, \frac{7}{13}, \frac{14}{19}, \frac{21}{32}, \frac{35}{51}, \frac{7}{12}$

looking at sequence numbers as ~~decimals~~ percentages (approximate)

100, 50, 66.6, 60, 85.7, 75, 62.5, 90.9, 78.9, 53.8, 73.6, 65.6, 68.6, 58.3,

it follows pattern of lower-higher except for 85.7-75-62.5 and 90.9-78.9-53.8. However the range did not seem to be diminishing.

Let's see if there's a process by which if $\frac{n}{x}$ is a sequence number, $\frac{n+1}{x}$ must also be

$\frac{3}{5}$: $\frac{1}{\frac{2}{3}+1}$ generated $\frac{3}{5}$

~~So for $\frac{3}{5}$ we'd need $\frac{1}{\frac{2}{3}+\frac{1}{2}}$~~

$$\frac{1}{\frac{3}{5}+x} = \frac{4}{5}$$

$$5 = \frac{12}{5} + 4x$$
$$\frac{13}{5} = 4x$$
$$x = \frac{13}{20}$$

→ Fibonacci looking

$\frac{21}{32}$

$$\frac{1}{\frac{21}{32}+x} = \frac{22}{32}$$

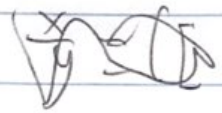
$$32 = \frac{462}{32} + 22x$$
$$\frac{562}{32} = 22x$$
$$\frac{562}{704} = x$$
$$\frac{281}{352} = x$$

→ no clear pattern

$$\frac{x}{y} = \left[\left[\left[(a+b)^{-1}+(a+b)^{-1}\right]^{-1} + (a+b)^{-1}\right]^{-1} + (a+b)^{-1}\right]^{-1} \cdots$$ pg 12

$$\frac{x}{y} = \left[\left[2(a+b)^{-1}\right]^{-1} + (a+b)^{-1}\right]^{-1} + (a+b)^{-1}\right]^{-1} \cdots$$

$$\frac{x}{y} = \left(\left[\tfrac{1}{2}(a+b) + (a+b)^{-1}\right]^{-1} + (a+b)^{-1}\right)^{-1} \cdots$$



I don't think this will show possible sequence numbers

Any fraction $\frac{x}{y}$ can be characterized as $\frac{n}{2}x$. That might not eventually lead back to a sequence number, however since there are infinite sequence numbers ($\frac{n}{x} \neq \frac{x}{n+x}$ is true since $x \neq 0$) there are an infinite number of distinct rules like this. It feels as though this indicates any fraction between $\frac{1}{2}$ and 1 is fashionable, although this reasoning does not seem concrete

$\frac{1}{1+\frac{1}{2}} = \frac{2}{3}$ $\quad \frac{1}{\frac{1}{1+\frac{1}{2}} + \frac{1}{1+\frac{1}{2}}} = \frac{3}{4}$ but if you plug in $\frac{3}{4}$ you just get back $\frac{2}{3}$ → can this be rewritten?

$$\frac{1}{\left(\frac{1}{\frac{1}{1+\frac{1}{2}}+1}+1\right)} = \frac{5}{8}$$

What I said with the rule of $\frac{1}{x}$ is that if x is a sequence number, $\frac{1}{x}$ is also a sequence number. What this is doing is saying that

Sequence of numbers generated by $\frac{1}{2}$:

$\frac{1}{2}$: 1, $\frac{2}{3}$, $\frac{6}{7}$, $\frac{14}{19}$, $\frac{38}{47}$, $\frac{94}{123}$, $\frac{246}{311}$

$\frac{2}{3}$: $\frac{3}{5}$, $\frac{6}{7}$, $\frac{3}{4}$, $\frac{12}{17}$,

This system of looking at each number and trying to find a pattern also does not account for $\frac{2}{3}$ being added to a number generated by, for example, $\frac{1}{2}$, and so doesn't even encompass a section of the set.

(of course this "section" would be infinitely smaller than the actual set)

What if the sequence was that if a and b are fashionable, a+b is fashionable instead of $\frac{1}{a+b}$?

1, 2, 3, 4, 5, 6, 7, 8, 9...

it just means all integers are fashionable. How does this change for $(a+b)^{-1}$?

It's not like you just make the former set into denominators of numbers like $\frac{1}{3}$ would be part of the sequence and $\frac{1}{3}$ & $\frac{1}{2}$

~~[illegible]~~

~~[illegible]~~

1 and 1 → 2 now 1 and 2 → $\frac{1}{2}$

2 and 2 → 3 now 1 and $\frac{1}{2}$ → $\frac{2}{3}$

2 and 2 → 4 now $\frac{1}{2}$ and $\frac{1}{2}$ → 1

2 and 3 → 5 now $\frac{1}{2}$ and $\frac{2}{3}$ → $\frac{6}{7}$

3 and 3 → 6 now $\frac{2}{3}$ and $\frac{2}{3}$ → $\frac{3}{4}$

3 and 4 → 7 now $\frac{2}{3}$ and 1 → $\frac{3}{5}$

4 and 4 → 8 now 1 and 1 → $\frac{1}{2}$

4 and 5 → 9 now 1 and $\frac{6}{7}$ → $\frac{7}{13}$

5 and 5 → 10 now $\frac{6}{7}$ and $\frac{6}{7}$ → $\frac{7}{12}$

5 and 6 → 11 now $\frac{6}{7}$ and $\frac{3}{4}$ → $\frac{28}{45}$

6 and 6 → 12 now $\frac{3}{4}$ and $\frac{3}{4}$ → $\frac{2}{3}$

greater than $\frac{1}{2}$

Every fraction^ can be expressed as 1 divided by the sum of two fractions greater than $\frac{1}{2}$ in an infinite amount of ways. I think that every fraction between $\frac{1}{2}$ and 1 will be fashionable then as each of the fractions expressing the fraction you are trying to claim is a fashionable number can also be expressed as 1 over the sum of two fractions greater than $\frac{1}{2}$ in an infinite amount of ways. Given that this is not concrete logic, I am definitely uncertain about this, but I think the set of fashionable numbers may be able to be expressed in the following way: pg 15

$\{x \mid x \in \mathbb{Q}, \frac{1}{2} \leq x \leq 1\}$

[redacted]

How would one

Is there a way to symbolically express the summation process of $\frac{1}{a+b}$

it has to start with $\frac{1}{1+1}$, or $\frac{1}{2}$, but at that point it bridges off because you could do $\frac{1}{1+\frac{1}{2}}$ or $\frac{1}{\frac{1}{2}+\frac{1}{2}}$ or $\frac{1}{1+1}$

$$\frac{1}{\frac{1}{\frac{1}{2}+a}+\frac{1}{b+c}} \quad a+c,\ a-c$$

$$\left[\left\{\left[(a+a)^{-1}+(b+c)^{-1}\right]^{-1}+(d+e)^{-1}\right\}^{-1}+(f+g)^{-1}\right]^{-1} \ldots$$

where a can be $\frac{1}{2}$ or 1, b and c can be any number that

You could also say that α and a will be equal to 2 since that is the only given fashionable numbers, however this is unnecessary and the variables are only needed when they in fact can represent more than one number

In the above equation, a and α can be equal to $\frac{1}{2}$ or 1, b and C can be equal to any quantity that can be made from $(a+\alpha)^{-1}$, with given a and α's possible values, b and C can also equal to 0, can just one of them equal 0? Actually I should not have written out the expression like I did on the previous page. It should not have two terms in each parenthesis, all should have one except the first one:

$$\left[\left\{\left[(\alpha+a)^{-1}+b\right]^{-1}+C\right\}^{-1}+d\right]^{-1}\ldots$$

α and a can be equal to either $\frac{1}{2}$ or 1, and α does not have to equal a, b can be equal any possible value of $[\alpha+a)^{-1}$, C can be equal to any possible value of $(\alpha+a)^{-1}$ or any value of $[(\alpha+a)^{-1}+b]^{-1}$ and this pattern continues. None of the numbers may equal 0, however any sum to the -1 is a fashionable number. In other words $(\ldots)^{-1}$ is a fashionable number using the variables above

Really this could be expressed as

$$X=(\alpha+a)^{-1},\ [(\alpha+a)^{-1}+b]^{-1},\ \{[(\alpha+a)^{-1}+b]^{-1}+C\}^{-1},\ldots$$

with the variables having the same value as above

So here is how the set of all fashionable could be expressed:

$\{X \mid X \in \mathbb{Q}, X=(\alpha x+a)^{-1}, X=[(\alpha x+a)^{-1}+b]^{-1}, X=\{[(\alpha x+a)^{-1}+b]^{-1}+(\quad)^{-1}, \ldots,$
$\alpha \neq \frac{1}{2}, 1, a \neq \frac{1}{2}, 1, b=(\alpha x+a)^{-1}, c=(\alpha x+a)^{-1}[(\alpha x+a)^{-1}+b]^{-1}, \ldots\}$

Below is, I believe, a much better answer than my prior answer. This set encompasses the ~~possible~~ fashionable numbers by using the process by which they are generated, while my other answer was little more than an instinct without solid logic behind it. Here's how the set of all fashionable numbers could be expressed:

$\{X \mid X=(\alpha x+a)^{-1}, X=[(\alpha x+a)^{-1}+b]^{-1}, X=\{[(\alpha x+a)^{-1}+b]^{-1}+(\quad)\}^{-1}, \ldots, X \in \mathbb{Q},$
$\alpha \neq \frac{1}{2}, 1, a \neq \frac{1}{2}, 1, b=(\alpha x+a)^{-1}, c=(\alpha x+a)^{-1}[(\alpha x+a)^{-1}+b]^{-1}, \ldots\}$

C) Word on the street is that in 2025, the fashion rules will be the same except for "1 is fashionable". Explore how the set of fashionable numbers with a different starting set of numbers. For example, what if 2 and $\sqrt{3}$ are the given fashionable numbers?

What happens if we start with just 2?
Will the domain be from $\frac{1}{4}, 2$?

$\frac{1}{2+2} = \frac{1}{4}$ $\frac{1}{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\frac{1}{2}} = 2$

$\frac{1}{2+\frac{1}{4}} = \frac{1}{\frac{9}{4}} = \frac{4}{9}$

The same logic applies. Since $\frac{1}{2+2}$ creates a number less than 2, and since $\frac{1}{2+a}$ where $a<2$ results in a fraction greater than $\frac{1}{4}$, $\frac{1}{4}$ must be the smallest number. The largest number will come from the smallest denominator, which means it will come from adding the smallest number in the set to itself. Since the smallest number in the set is $\frac{1}{4}$, the largest number will be equal to $\frac{1}{\frac{1}{4}+\frac{1}{4}} = \frac{1}{\frac{1}{2}} = 2$. Therefore the domain of the set of fashionable numbers where 2 is the starting number is $[\frac{1}{4}, 2]$

Adding $\sqrt{3}$ to the given numbers shouldn't change anything, as the domain of the set will be determined by the largest given number. The logic above still applies as $\sqrt{3}$ is less than 2. However, $\sqrt{3}$ does mean that all the numbers in this set are no longer rationals, as $\sqrt{3}$ is irrational (shown in problem 1 part a).

However, I was incorrect to say that the largest number in the set determines the domain- you must first raise all numbers less than 1 to the -1 power and then divide by two and see which number is the greatest. That number will determine the domain

ex) given numbers: 2, $\frac{1}{5}$ $\quad (\frac{1}{5})^{-1} = 5 \rightarrow 5/2 = 2.5$

2.5 > 2. $\frac{1}{5}$ will determine the domain of the set, $\frac{1}{5}$ will be the smallest number and $\frac{1}{\frac{1}{5}+\frac{1}{5}}$ will be the largest number (if the number is greater than 1 it will be the largest of the set and 1 divided by twice its value will be the smallest)

$$\frac{1}{\frac{1}{5}+\frac{1}{5}} = \frac{1}{\frac{2}{5}} = \frac{5}{2}$$

$\frac{5}{2} > 2$

Let's see if induction can be used, but this time instead of 1 being the given number, the variable x will represent the given number. This will apply then to all situations with 1 given number. If there is more than one given number, see above in terms of which number determines domain.

x = given number (x > 1)
(if x is less than 1, do $\frac{1}{2x}$ and use that number)

Claim: $\frac{1}{2x} \leq \frac{1}{a+b} \leq X$

P(1): $\frac{1}{2x} \leq \frac{1}{x+x} \leq X$

$\frac{1}{2x} \leq \frac{1}{2x} \leq X$ True

assume $\frac{1}{2x} \leq a \leq X$ and $\frac{1}{2x} \leq b \leq X$

$\frac{1}{2x} \leq \frac{1}{a+b} \leq X$

$\frac{1}{x} \leq a+b \leq 2x$

$a+b \geq \frac{1}{x}$ $\quad$ $a+b \leq 2x$

Since this is saying a+b has to be greater than or equal to a quantity, let's look at the smallest possible value of a+b

Since this is saying a+b has to be less than or equal to a quantity, let's look a the greatest possible value of a+b

$\frac{1}{2x} + \frac{1}{2x} \geq \frac{1}{x}$ $\quad$ $X+X \leq 2x$

$\frac{2}{2x} \geq \frac{1}{x}$ $\quad$ $2x \leq 2x$

$\frac{1}{x} \geq \frac{1}{x}$ $\quad$ $2x \leq 2x$

True $\quad$ True

Therefore we can conclude by mathematical induction that the range of $[\frac{1}{2x}, X]$ is valid for every number generated after X by this sequence (when X is given number)

Using the previous proof and the rule about the largest number, after raising all numbers less than 1 to the -1 power and dividing by 2, we can first determine that if the numbers 2 and $\sqrt{3}$ are given that 2 will be the number controlling the domain (2 > $\sqrt{3}$, $\sqrt{3}$ > 1 so no need to raise to -1 and divide by 2) and that the domain will be from $\frac{1}{4}$ to 2 inclusive.

Furthermore, we can consider the function of $\frac{1}{a+b+c}$ or $\frac{1}{a+b+c+d}$. Here the same logic that the largest number will be created by the addition of the smallest fraction, and the smallest number by the largest numbers being added to itself however many times. However the way to determine the given number that is responsible for the domain restrictions changes:

instead of ~~[illegible]~~ taking numbers that are less than one, raising them to the -1 power and dividing by 2, you will raise them to the -1 power and divide by the amount of terms in the denominator.

Ex) if you have the function $\frac{1}{a+b+c}$ and the given numbers 2 and $\frac{1}{5}$

if you do $[(\frac{1}{5})^{-1}]\frac{1}{2}$ you'll get $\frac{5}{2}$ which is greater than 2, however if we look at the numbers generated by the function we see that this is wrong:

$\frac{1}{2+2+2} = \frac{1}{6}$ $\frac{1}{6} < \frac{1}{5}$ $\frac{1}{\frac{1}{5}+\frac{1}{5}+\frac{1}{5}} = \frac{1}{\frac{3}{5}} = \frac{5}{3}$ $\frac{5}{3} < 2$

So we must raise $\frac{1}{5}$ to the -1 and divide by 3 (since there are three terms. The more terms the larger the numerator of the fraction gets and the numerator becomes the denominator. So, this must be accounted for by dividing the the fraction's denominator, which will become it numerator by its numerator, ~~[illegible] amount of terms equals numerator since function has numerator of 2~~ which will become its denominator

~~[illegible]~~ also in this case the ~~domain is~~ range is between $\frac{1}{6}$ and $\frac{2}{3}$, which is between $[\frac{1}{3x}, X]$

So the following is the range of numbers of the function of $\frac{}{a+b+c+d...}$ with n given numbers

Let X represent the number that determines the range of the function CX can be determined as priorly demonstrated

Let y represent the number of terms in the denominator

Let S represent the above function of $\frac{1}{a+b+c+d...}$

The domain of S is $[\frac{1}{xy}, X]$

Finally, for the function S of $\frac{1}{a+b+c+d...}$, where X represents the number that determines the range, and where ~~[illegible]~~ equals all given values (original numbers given)

$\{X \mid X=(a+b+c+d...)^{-1}, X=[(a+b+c+d...)^{-1}+\alpha+\beta+\gamma...]^{-1},$
$X=\{[(a+b+c+d...)^{-1}+\alpha+\beta+\gamma...]^{-1}+\lambda+\nu+\zeta...\}^{-1}...\}$
$a,b,c,d...=f, \alpha,\beta,\gamma...=f, \alpha,\beta,\gamma...=(a+b+c+d...)^{-1}\}$
$\zeta,\nu,\zeta...=f, \lambda,\nu,\zeta...=(a+b+c+d)^{-1}, \lambda,\nu,\zeta=[(a+b+c+d...)^{-1}+\alpha+\beta+\gamma...]^{-1}$
$...\}$

denominator terms

I used different alphabets just because the sequence could be any length.

The above set is not necessarily an element of the rationals, or even the real numbers because it depends on the given number(for example i could be a given number). Additionally it is made so that after the first term is created, the $\frac{1}{a+b+c+d...}$ any term could represent any of the given or already generated terms. As more terms are generated, the variable representing the later terms have more possible values.

4. We'd like for you to define the squarilarity of two-dimensional shapes. The squarilarity should be a number between 0 and 1 quantifying how much a shape looks like a square. All squares should have squarilarity equal to 1, but aside from that restriction you can decide how to assign numbers to various shapes. We welcome and are interested in seeing any ideas you have. It may be a good idea to start thinking of squarilarity definitions for simple shapes, and then expanding to more complicated shapes. Ultimately, for whatever definition(s) you come up with, we'd like for you to the following

A) Specify the class of shapes for which your definition applies

B) State your definition of squarilarity for the shapes you're considering

C) Verify that the squarilarity is always between 0 and 1, and verify that the squarilarity of any square is 1.

D) Explore the implications of your definition! Compute the squarilarity of a few shapes, and make observations of patterns you notice. Is your definition of squarilarity a good one?

The first definition will only apply to rectangles

$S = \frac{W}{L}$ S=squarilarity W=width L=length

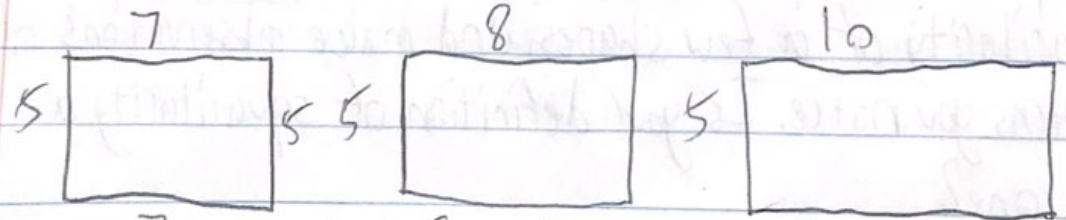
In squares the width and length are the same, so the squarilarity would be 1:

$S = \frac{W}{L} \rightarrow S = \frac{L}{L} \rightarrow S = 1$

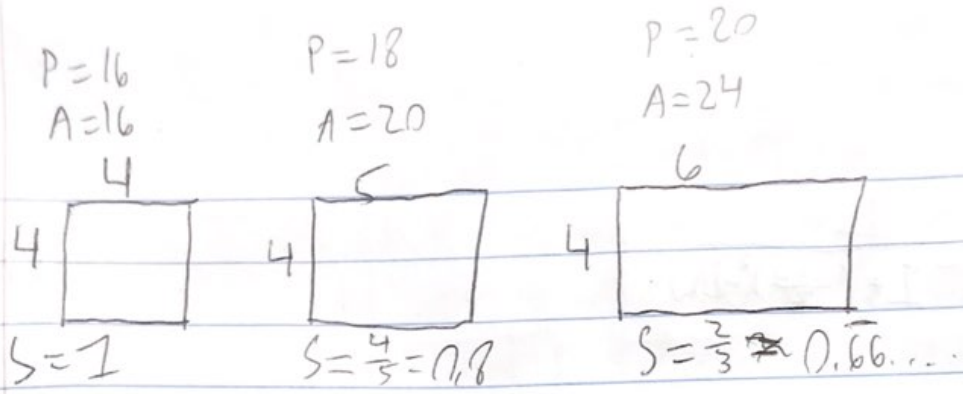
Since width refers to the shorter side of a rectangle, all values of squarilarity would be less than or equal to 1, and since the definition refers to ~~two-dimensional~~ rectangles, which are two-dimensional, the width must be greater than 0. Hence the squarilarity of all rectangles must be between 0 and 1.

In this case, the definition of squarilarity is how similar in value the width and length are, particularly what percentage of the length is the width

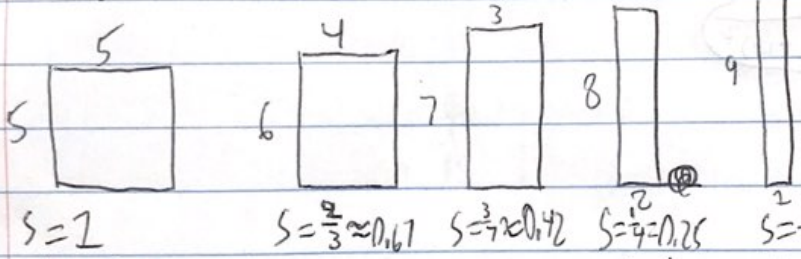
examples:



$S = \frac{5}{7} \approx 0.71$ $S = \frac{5}{8} \approx 0.63$ $S = \frac{1}{2} = 0.5$



The difference in squarilarity between a 4×4 and a 4×5 is greater than the difference between a 4×5 and a 4×6, which I think is good because if you look at a rectangle the difference between something like a 4×4 and a 4×5 is much more prominent than the difference between a 4×100 and a 4×101



for rectangles

Another definition could be to average the width and the length, square that quantity, and subtract it from the value of the width times the length (then you also get squarilarity in terms of proximity to closest square, and squarilarity would correspond to area, however the values wouldn't be between 0 and 1, however to fix that you could do ~~1 the above quantity~~ 1 over 1+ the above quantity

$$S = \frac{1}{1+\left(\frac{L+W}{2}\right)^2 - LW}$$

Can this be simplified?

$$\frac{1}{1+\left(\frac{L+W}{2}\right)\left(\frac{L+W}{2}\right) - LW} = S$$

$$\frac{1}{1+\frac{L^2+W^2+2LW}{4} - LW} = S$$

$$S = \frac{4}{4+L^2+W^2-2LW}$$

$$S = \frac{4}{4+(L-W)^2}$$

$$S = \frac{1}{1+\frac{1}{4}(L-W)^2}$$

$$S = \frac{1}{1+\left[\frac{1}{2}(L-W)\right]^2}$$

→ this looks more simplified

The squarilarity of all squares are 1 under this definition,

$L = W$ for squares

$$S = \frac{4}{4+(L-L)^2} \rightarrow S = \frac{4}{4+0^2} \rightarrow S = \frac{4}{4} \rightarrow S = 1$$

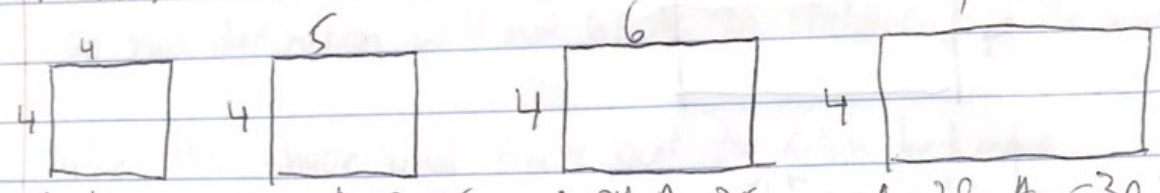
Since $(L-W)^2$ can never be negative, since $L \geq W$, $\frac{4}{4+(L-W)}$ must always be equal to or less than 1 since the denominator must be equal to or greater than 4

$S = \frac{4}{4 + (L-W)^2}$

*A_S = Area of square using the average of the length and width

pg 5

Additionally, since the numerator is 4, the squarilarity must not equal 0. Hence, the squarilarity must be between 0 and 1, and can't be equal to 0.



$A = 16$ $S = 1$

$A = 20$ $A_S = 20.25$ $S = \frac{4}{5} = 0.8$

$A = 24$ $A_S = 25$ $S = \frac{1}{2} = 0.5$

$A = 28$ $A_S = 30.25$ $S = \frac{4}{13} \approx 0.31$

This definition is a bit odd because as the length increases by intervals of 1, the difference in squarilarity first gets larger and then gets smaller, so this definition is likely not as good as the first one.

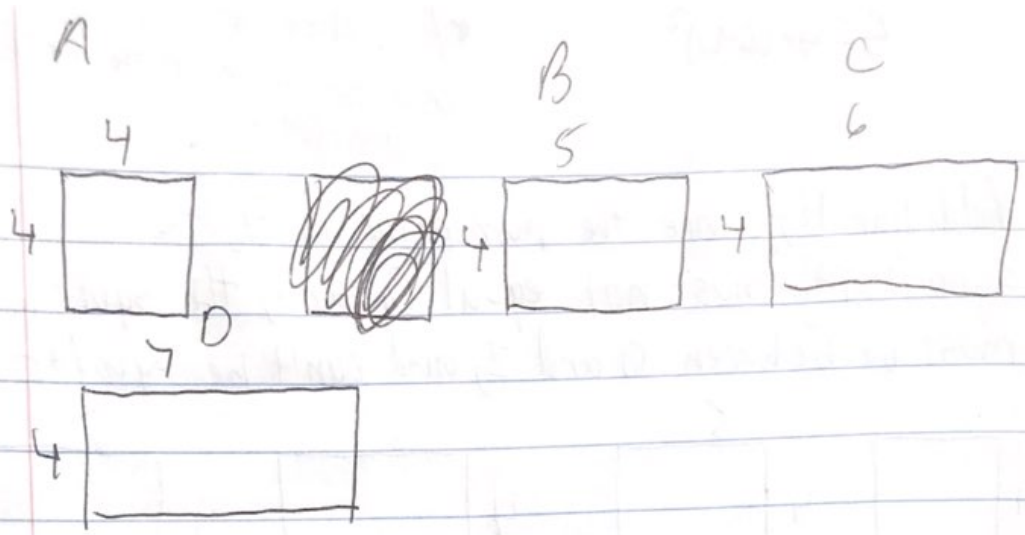
This definition of squarilarity is about how close each rectangle's area is to the area of the square using the average sides of the rectangle

What about looking at the first definition of squarilarity again, but squaring the quantity of $(\frac{W}{L})$

$S = (\frac{W}{L})^2$

$[0,1]$

Squarilarity must still be between 0 and 1, as the width is never negative nor 0, and the length is always greater than or equal to the width ($L \geq W$)



pg 6

A: $S = (\frac{4}{4})^2 = 1$

B: $S = (\frac{4}{5})^2 = \frac{16}{25} = 0.64$

C: $S = (\frac{4}{6})^2 = \frac{16}{36} = \frac{4}{9} = 0.\overline{44}$

D: $S = (\frac{4}{7})^2 = \frac{16}{49} \approx 0.33$

In this definition the squarilarity of all squares is still 1.

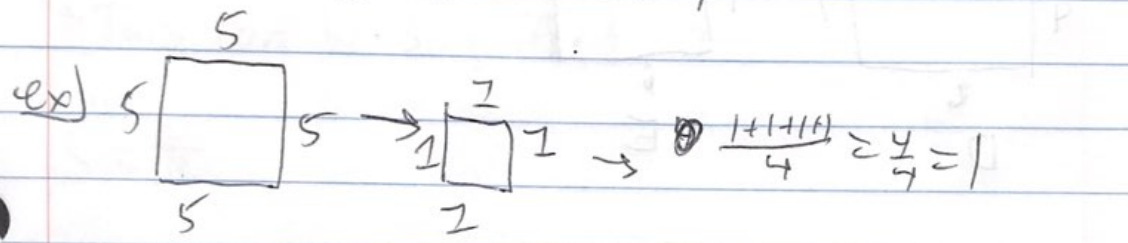
$$S = (\frac{W}{L})^2 = (\frac{L}{L})^2 = 1^2 = 1$$

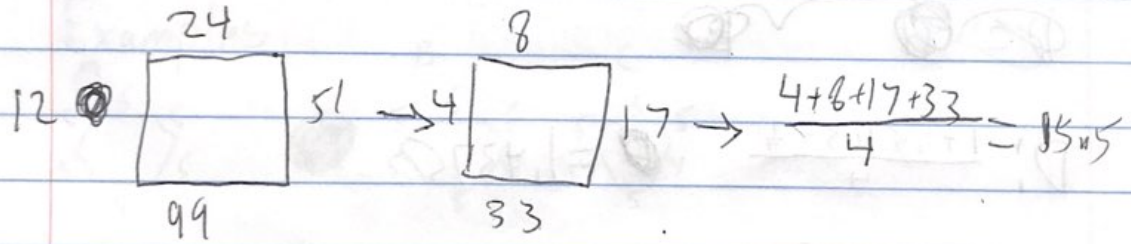
The rate of change still decreases as the dimensions get further and further away from a square, and here the rate of change of the rate of change is larger

This is also a decent definition of squarilarity, however I still believe the 1st one to be the best, as the Squarilarity value of a 4×5 compared to a 4×4 is very high, 0.36, whereas the difference between a 4×6 and 4×7 is only 0.11, which I believe is too much of a difference from 0.36

For the next definition, I will expand the class of shapes applicable to all two dimensional shapes with an natural number N of sides (so this definition will not apply to circles, ellipses, the like)

take the shape and figure out the ratios between the sides, and add those numbers, and divide by the amount of sides of the shape

ex) 

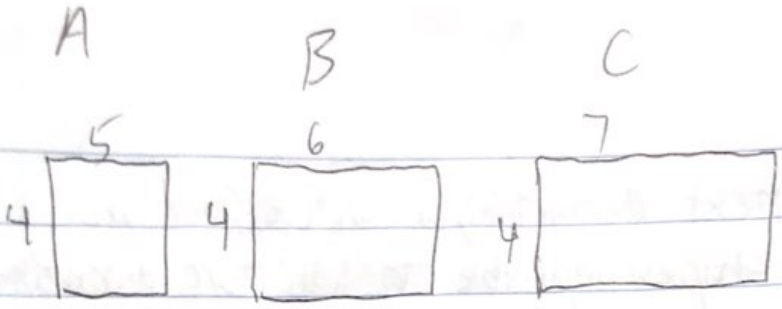


There is a problems with this method though in

~~It doesn't account for the difference in each side, so a 2x5 rectangle~~

that the squarularity can be greater than 1.

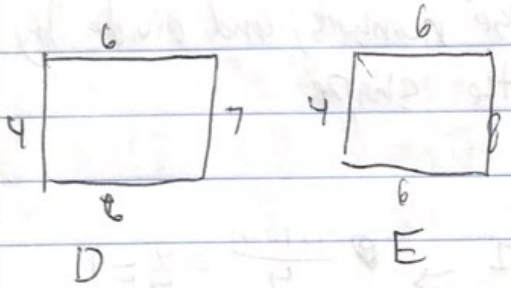
Also a 4x7 would be considered further from a square than a 4x8, even though it's not. Maybe writing all ratios in ~~terms of~~ with the lowest unit of 1 would solve this



A: $\frac{1+1+1.25+1.25}{4} = 1.125$

B: $\frac{1+1+1.5+1.5}{4} = 1.25$

C: $\frac{1+1+\frac{3}{4}+\frac{3}{4}}{4} = 1.375$



D: $\frac{1+1.5+1.5+\frac{3}{4}}{4} = 1.4375$

E: $\frac{1+1.5+1.5+2}{4} = 1.5$

Now the definition's more consistent, however the values still range from ~~greater than~~ anything between 0 and infinity (not inclusive) $(0, \infty)$

To fix this make the sum of the ratios, with the smallest side as one, divided by the amount of sides of the shape the denominator and make the numerator equal to 1

$S = \frac{1}{(\frac{R}{N})}$

S = squarilarity

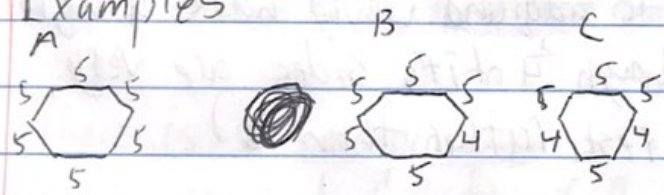
R = sum of ~~[illegible]~~ of all sides when the smallest side is set equal to one and the ratios between sides are maintained

N = number of sides of the shape

→ This can be simplified to

$S = \frac{N}{R}$

Examples



$A = \frac{6}{6} = 1$ → This problem will be addressed later, likely by adding a second component based on amount of sides

$B = \frac{6}{1+5(1.25)} = \frac{6}{7.25} = \frac{24}{29} \approx 0.83$

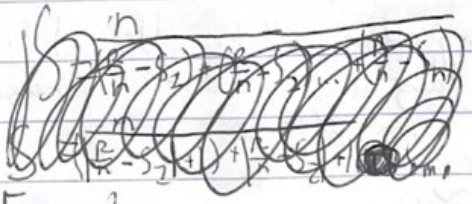
$C = \frac{6}{2+4(1.25)} = \frac{6}{7} \approx 0.86$

This is a problem because you could have an octagon with every side being 100 except one is 1 unit, and an octagon where 6 sides are 100 units and 2 sides are 1 unit, and the second octagon has a higher squarilarity value

* S must be greater than 0 since N>0, S must be less than or equal to one, since absolute value is always positive or 0, so the denominator must at least equal N, and $\frac{N}{N}=1$, Additionally since every side is the average side length for a square, $S=\frac{N}{n}=1$, so all squares have a squarilarity of 1

n/10

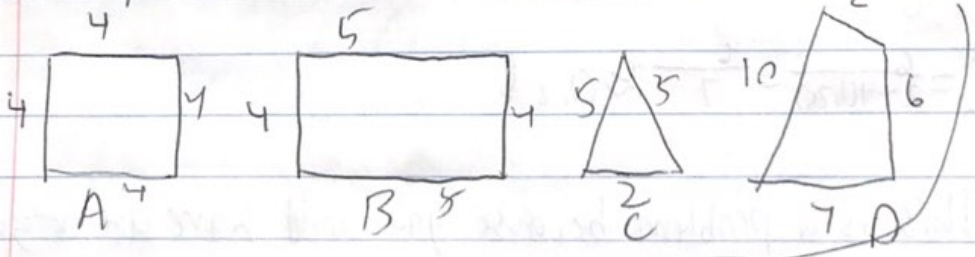
This problem can be fixed by ~~calculating~~ taking the sum of all the sides (not ratioed, actual lengths) and dividing by the amount of sides, and then figuring out the difference between the average and each side (in absolute value) and finding the sum of that quantity, and setting that value as the denominator. The numerator will stay as the amount of sides (because shapes with more sides have more numbers to add, so shapes like the octagon with 7 sides as 100 and 1 side as 2 are closer to being even in their sides than shapes like a 4 sided shape with dimensions 100×100×100×2, however the latter shape has less numbers to add, and would have a higher squarilarity even though $\frac{1}{4}$ of its sides are very different from the rest rather than $\frac{1}{8}$)

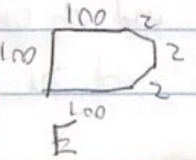


n = amount of sides
R = sum of all sides
S_1 = side 1

$$S=\frac{n}{(|\frac{R}{n}-S_1|+1)+(|\frac{R}{n}-S_2|+1)\ldots(|\frac{R}{n}-S_n|+1)}$$

Examples

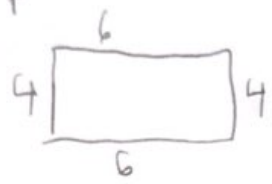




→ Another way of writing this is

$$S=\frac{n}{|\frac{R}{n}-S_1|+|\frac{R}{n}-S_2|\ldots|\frac{R}{n}-S_n|+n}$$

F



A: $S = \frac{4}{\left|\frac{16}{4}-4\right|+1+\left|\frac{16}{4}-4\right|+1+\left|\frac{16}{4}-4\right|+1+\left|\frac{16}{4}-4\right|+1} = \frac{4}{0+1+0+1+0+1+0+1} = \frac{4}{4} = 1$

B: $S = \frac{4}{|4.5-4|+|4.5-4|+|4.5-5|+|4.5-5|+4}$

$S = \frac{4}{6} = \frac{2}{3} \approx 0.66$

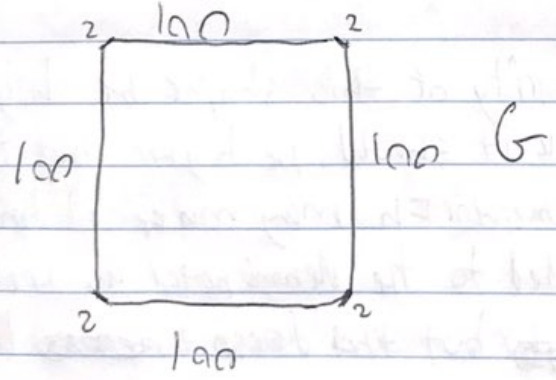
C: $S = \frac{3}{|4-5|+|4-5|+|4-2|+3} = \frac{3}{7} \approx 0.43$

D: $S = \frac{4}{|6.25-10|+|6.25-7|+|6.25-6|+|6.25-2|+4} = \frac{4}{13} \approx 0.31$

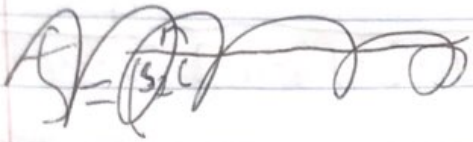
E: $S = \frac{6}{|51-100|+|51-100|+|51+100|+|51-2|+|51-2|+|51-2|+6} = \frac{6}{300} = 0.02$

F: $S = \frac{4}{|5-6|+|5-6|+|5-4|+|5-4|+4} = \frac{4}{8} = \frac{1}{2} = 0.5$

There is still a problem with this definition, which is illustrated by the following shape



This shape is nearly square, yet since all sides are weighted equally under the current definition of squarularity, its squarularity value would be very small



instead of calculating the average side by $\frac{P}{n}$, it will be calculated by $\frac{s_1^2+s_2^2 \ldots s_n^2}{P}$

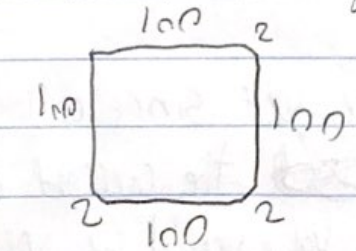
With this method, the average side length of the figure from before goes from 51 to 98.07. However, the difference between 98.07 and 2 is substantial, so maybe the summation of sides as part should be weighted as well

$$S = \frac{n}{\left(\frac{s_1|k-s_1| + s_2|k-s_2| \ldots + s_n|k-s_n|}{P}\right)+n}$$

S = side
P = perimeter
$k = \frac{s_1^2+s_2^2+\ldots+s_n^2}{P}$

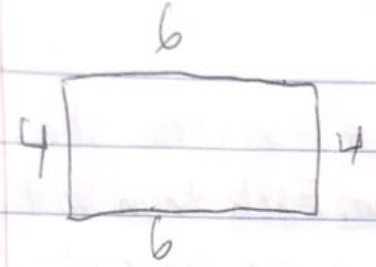
$$S: \frac{8}{\left(\frac{100\left|\frac{40,016}{408}-100\right|\cdot 4 + 2\left|\frac{40,016}{408}-2\right|\cdot 4}{408}\right)+8} \approx 0.68$$

Will the squarilarity of this shape be higher or lower? I think it should be higher but the fact that the numerator = n may make it lower, although n is added to the denominator as well, ~~it should be higher~~ but that doesn't balance it



$S \approx 0.74$

$k = 5.2$



pg 13

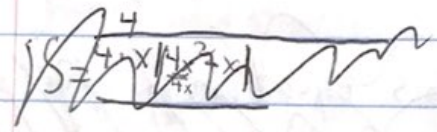
$S \approx 0.81$

I think this is actually a fairly decent definition of squarilarity, except for the fact that smaller sides still have to big an influence on the averages and that there is no accounting for sides of shape

Squarilarity is still always between 0 and 1, as $n > 0$ and the denominator consists of n + absolute value terms being multiplied and divided by positive numbers so the denominator must be greater than or equal to the numerator.

Furthermore squarilarity of squares does always equal 1 for squares!

x = side length



Let's just look at the term $|k - S_4|$

$|k - S_4|$ will be the same for every side, since they are all the same length sides

$k = \frac{4x^2}{4x}$ → four times each side squared / 4x — Perimeter

$$\left|\frac{4x^2}{4x} - x\right| \to |x - x| \to 0$$

you'll end up with $\frac{n}{n+0} = \frac{n}{n} = 1$

To incorporate the amount of sides into the right calculation, I will add a term. Each term will be multiplied by $\frac{1}{2}$ so that the range of $[0, 1]$ stays

n = number of sides of shape

For 3 sided shapes: $\frac{n}{4} = S$

For all other shapes: $\frac{4}{n} = S$

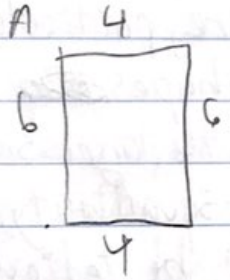
This does mean that a shape with 3 sides is considered farther from a square than a triangle, although that kind of does make sense as percentage wise the S is closer, and often the actual shapes look closer, though of course it would depend on the shape. Either way it does seem reasonable to me

Another nice thing about this equation is that when you input 1 as n you get 0, which is good since there is no 2 dimensional shape that has 2 sides, as there is no way to make a bounded area, so they are not included in the above definitions of squarishness

* For objects with more than 3 sides change the $\frac{n}{4}$ to a $\frac{4}{n}$

Also, the squarilarity of 4 sided shapes would be 1 pg 15 under this definition

The prior equations have the same problem with not being weighted averages ~~and so should be adapted to the following~~ but if you add a weighted average shapes like A end up essentially ~~being~~ getting subtracted twice because of their dimensions, not the amount of sides



So the equations won't be modified, and this definition can be applied to shapes with greatly different side lengths, however the result may not be as good as for shapes with side lengths closer in length.

For 3 sided shapes:

$$\int = \frac{1}{2}\left(\frac{n}{4}\right) + \frac{1}{2}\left(\frac{n}{\frac{s_2|k-s_2| + s_2|k-s_2| \ldots + s_n|k-s_n|}{P} + n}\right)$$

n = amount of sides of shape

P = perimeter

$k = \frac{s_2^2 - s_2^2 \ldots + s_n^2}{P}$

Since I have already demonstrated that the squarilarity of squares under each of the previous individual definitions, multiplying each by $\frac{1}{2}$ and then adding the two terms must also result in 1 as $\frac{1}{2} + \frac{1}{2} = 1$

Similarly since neither term can be 0 or negative their sum must be greater than 0

Since both terms were not able to reach a value greater than $\frac{1}{2}$, each term individually can now reach a maximum of $\frac{1}{2}$. Since $\frac{1}{2}+\frac{1}{2}=1$ the Squarilarity must be between 0 and 1 $[0,1]$.

This definition is a good one, particularly in that it applies to so many shapes and uses a weighted average so that the longer sides are more important to the objects squarilarity, however it is complicated and it can be tedious to compute. So, using the same principals of comparing side lengths and amount of sides, here is a simpler definition, and I believe a better one

r=sum of all sides
perimeter

$\frac{1}{n}$

This is, I believe, a more logical version of the length comparison term pg 17

$$\frac{1}{\frac{1}{n}\left(\left|\frac{r}{n}-S_1\right|+\left|\frac{r}{n}-S_2\right|\ldots+\left|\frac{r}{n}-S_n\right|\right)+1}$$

r = perimeter
n = amount of sides

~~This does not apply to squares themselves which must use the alternate equation~~

Before n was the numerator, and I tried to balance that to same degree by adding n to the denominator but ~~not~~ this doesn't really work

In this equation the average difference from the average side length is calculated, by adding the difference between the average side length and each side, and that total is divided by the amount of sides to determine the average distance from the average side length.

The use of absolute value terms means the differences don't just cancel out, and squarilarity will always be between 0 and 1 because the numerator is never 0 or negative, nor are any terms negative. Also the denominator is always greater than or equal to 1 as it is 1 plus a positive term times a positive number ($\frac{1}{n}$). Additionally, the squarilarity of all squares is 1 because $\frac{r}{n}=S_1=S_2=S_3=S_4$, so the summation adds to 0, $\frac{1}{4}=0$ $0+1=1$ and $\frac{1}{1}=1$

* you should take out common factors before you begin calculation (an 8×8×12×12 would become a 4×4×6×6). This is done so if you actually have a shape you can't just change units, for irrational numbers, since you can't multiply it by a rational number to get an integer, just set your lowest length to 1 ~~sides need to be expressed~~

↳ and maintain ratios

It is also simpler as there is no squaring necessary nor are there any terms that need to be ~~weighted~~ based on side length. All that needs to be done now is to combine it with the term that accounts for the amount of sides of the shape

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if $n=3$ (or $n=4$) ~~$n \in \mathbb{N}$~~ $n \in \mathbb{N}$

$$\int = \frac{1}{\frac{1}{n}\left(\left|\frac{r}{n}-S_2\right|+\left|\frac{r}{n}-S_2\right|_{...}+\left|\frac{r}{n}-S_n\right|\right)+1}\left(\frac{1}{2}\right)+\frac{n}{4}\left(\frac{1}{2}\right)$$

if $n>3$

$$\int = \frac{1}{\frac{1}{n}\left(\left|\frac{r}{n}-S_2\right|+\left|\frac{r}{n}-S_2\right|_{...}+\left|\frac{r}{n}-S_n\right|+1\right)}\left(\frac{1}{2}\right)+\frac{4}{n}\left(\frac{1}{2}\right)$$

r=perimeter
n=number of sides

This applies to 2d shapes that don't include curves

This is I believe the best of my definitions of squarilarity, taking into account how close the side lengths are to the average side length, and the amount of sides of the shape in a way that is not overcomplicated and has clear logic to it.

The proximity of side lengths to the ~~average~~ average side lengths and the amount of sides of the shapes are weighted equally (each multiplied by $\frac{1}{2}$ and then added)

Algebraic proof of equality of squares being equal to 1 with this definition:

$$S = \frac{1}{2}\left(\frac{1}{\frac{1}{4}(|4-4|+|4-4|+|4-4|+|4-4|+1)}\right)+\left(\frac{4}{4}\right)\left(\frac{1}{2}\right)$$

$$S = \frac{1}{2}\left(\frac{1}{\frac{1}{4}(0)+1}\right)+\frac{1}{2}(1)$$

$$S = \frac{1}{2}(1)+\frac{1}{2}(1)$$

$$S = \frac{1}{2}+\frac{1}{2}$$

$$S = 1$$

Additionally 0 is the smallest possible value for an absolute value term, and $0 \cdot \frac{1}{4} = 0$, and $0+1 = 1$ and $\frac{1}{1} = 1$ and $\frac{1}{2}(1) = \frac{1}{2}$. The largest value for $\frac{4}{n}$ where $n \in \mathbb{N}$ and $n > 3$ is 1, as 4 is the smallest possible value for the denominator. $(\frac{1}{2})(1) = \frac{1}{2}$, and so the sum of the terms adds to a maximum of $\frac{1}{2}+\frac{1}{2} = 1$

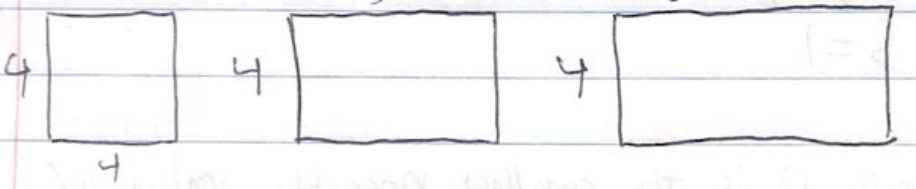
There are also no negative terms, so ~~the sum~~ S cannot be less than 0. S cannot equal 0, as the term $\frac{4}{n}$ requires that n be greater than 3, and so must be positive, and if the term $\frac{n}{4}$ is being used $n = 3$, making $\frac{n}{4}$ positive. $\frac{1}{2}$ times a positive number is still positive, so the second term must be positive. The first term must be positive because the absolute value terms times $\frac{1}{4}$ is either positive or 0. When you add 1, the number becomes positive, and 1 divided by a positive number is positive

Finally $\frac{1}{2}$ times a positive number is positive so term one must also be positive. Two positive terms added together equals a positive number, so S cannot be 0.

Domain of S: $(0, 1]$

Here are some examples of this ~~shape~~ definition of squarilarity:

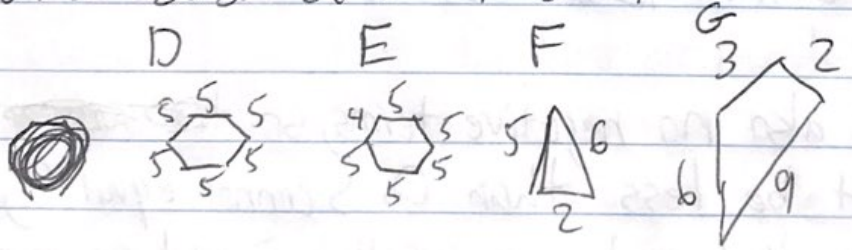
A



A: $\frac{1}{2}\left(\frac{1}{\frac{5}{4}[|4-4|+|4-4|+|4-4|+|4-4)+1}\right)+\frac{1}{2}\left(\frac{4}{4}\right)=1$ $S=1$

B: $S=\frac{1}{3}+\frac{1}{2}=\frac{5}{6}\approx 0.83$

C: $S=\frac{1}{2}\left(\frac{1}{2}\right)+\frac{1}{2}(1)=\frac{1}{4}+\frac{1}{2}=\frac{3}{4}=0.75$



D: $S=\frac{1}{2}(1)+\frac{4}{6}\left(\frac{1}{2}\right)=\frac{1}{2}+\frac{4}{12}=\frac{10}{12}=\frac{5}{6}\approx 0.83$

E: $S=\frac{1}{2}\left(\frac{1}{\frac{23}{18}}\right)+\frac{1}{2}\left(\frac{4}{6}\right)\approx 0.72$

$F: \frac{1}{2}\left(\frac{1}{\frac{1}{3}\left(\left|4\frac{1}{3}-5\right|+\left|4\frac{1}{3}-6\right|+\left|4\frac{1}{3}-2\right|\right)+1}\right)+\frac{1}{2}\left(\frac{3}{4}\right)$

$\frac{1}{2}\left(\frac{1}{\frac{1}{3}\left(\frac{13}{3}\right)+1}\right)+\frac{1}{2}\left(\frac{3}{4}\right)$

$\frac{1}{2}\left(\frac{1}{\frac{23}{9}}\right)+\frac{1}{2}\left(\frac{3}{4}\right)$

$S=\left(\frac{1}{2}\right)\left(\frac{9}{23}\right)+\frac{3}{8}$

$S=\frac{9}{46}+\frac{3}{8}\approx 0.57$

$G: \frac{1}{2}\left(\frac{1}{\frac{1}{4}\left(|5-3|+|5-2|+|5-6|+|5-9|\right)+1}\right)+\frac{1}{2}\left(\frac{4}{4}\right)$

$S=\frac{1}{2}\left(\frac{1}{\frac{1}{4}(2+3+1+4)+1}\right)+\frac{1}{2}(1)$

$S=\frac{1}{2}\left(\frac{1}{\frac{1}{4}(10)+1}\right)+\frac{1}{2}$

$S=\frac{1}{2}\left(\frac{1}{\frac{14}{4}}\right)+\frac{1}{2}$

$S=\frac{1}{2}\left(\frac{4}{14}\right)+\frac{1}{2}$

$S=\frac{4}{28}+\frac{1}{2}$

$S=\frac{1}{7}+\frac{1}{2}$

$S=\frac{2}{14}+\frac{7}{14}$

$S=\frac{9}{14}\approx 0.64$

This definition meets the requirements of providing a value between 0 and 1 and squares have a value of 1, and [redacted] is broad and seems logical to me.